

Pharmacology of Antineoplastic Agents Part 2

Targeted Anticancer Therapeutics

Protein Kinase Inhibitors

Pathway Targeted Small Molecules

Immune Checkpoint Inhibitors

CAR T Cells

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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Following this session, students will be able to:

1. Compare and contrast the actions and effects of the targeted anticancer drugs that block specific molecules in oncogenic pathways with immunotherapeutics that trigger the immune system to identify and attack cancer cells.
2. Compare and contrast the mechanisms of resistance of the targeted drugs and the conventional chemotherapies.
3. Relate the toxicities of the targeted agents with their mechanisms of action and effects.
4. Apply the mechanistic and pharmacokinetic properties of the targeted agents to their drug interactions.
5. Create a concept map or other technique integrating the mechanisms of action and resistance, potential toxicities, drug-disease and drug-drug interactions to the selection of drugs for combination antineoplastic therapeutic regimens.

Preparation Materials (links are in the CPG and on the next slide)

Required

- ScholarRx Bricks | Practice Questions (questions FOR learning)

Optional materials:

- Video Lecture | Dr. Goldstein's Notes handout | Guided Reading Questions
- Textbook resources with links are listed on the next slide

SUGGESTIONS:

- Use the resources that work best for you.
- You do not need to study all of them.
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Scholar Rx Bricks:

Name of Discipline/ Organ System: General Pharmacology > Chemotherapy > Chemotherapy Agents

Title of Brick: Alkylating and Platinum Agents

Link: https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_PHR0016

Title of Brick: Antimetabolite Agents

Link: https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_PHR0017

Title of Brick: Topoisomerase Inhibitors, Microtubule Inhibitors, and Antitumor Antibiotics

Link: https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_PHR0019

Access Medicine Katzung Basic & Clinical Pharmacology, 16e, 2024; Chapter 54: Cancer Chemotherapy

<https://nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/book.aspx?bookid=3382#281742356>

THE REVIEW BOOKS BELOW HAVE PRACTICE QUESTIONS

LWW Health Library, Medical Education; Lippincott's Illustrated Reviews, 8e, 2023; Chapter 37: Anticancer Drugs

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=253329619&bookid=3222>

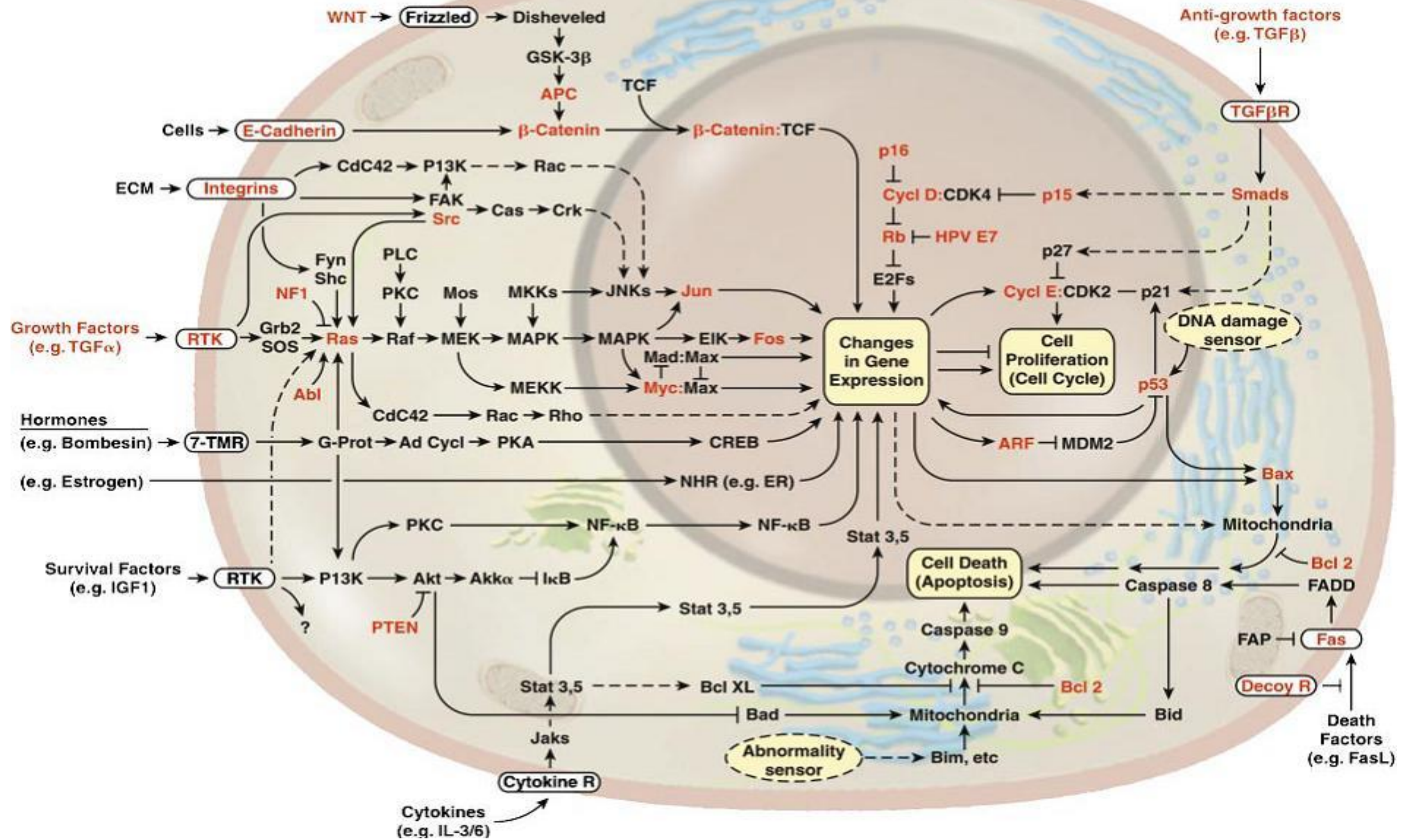
Access Medicine Katzung's Pharmacology: Examination & Board Review, 14e, 2024; Chapter 54: Cancer Chemotherapy

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461§ionid=285600220>

Approach to cancer treatment

Cancer treatment has evolved since the recognition of cancer genetics and the development of targeted therapeutics that specifically attack cancer cells.

- The drivers of cancer growth are the oncogenic pathways in malignant cells, the tumor microenvironment, and malignant cells' escape from host immune surveillance.
- Targeted anticancer therapeutics specifically act on well-defined extracellular or intracellular molecules to interrupt the oncogenic signaling, tumor growth, and proliferation.
- The targeted therapies are usually used in combination regimens including with nonselective cytotoxic chemotherapies and/or other targeted drugs.



Targets of biologic and small molecule targeted agents:

Targeted agents block specific molecules in oncogenic pathways that lead to dysregulated cancer cell growth and survival, which interferes with growth, progression, and spread of cancer.

Small molecule drugs and Biologics

- Small molecule drugs block intracellular targets, preventing activation of signaling pathways.
- Antibody therapeutics block soluble growth factors or specific cell surface receptors, preventing activation of the receptor's signaling pathway.
- Antibody-drug conjugates consist of an antibody that carries a cytotoxic payload to tumors overexpressing specific growth factor receptors.
- Immune checkpoint inhibitors unleash the immune system against the tumors.
- CAR-T cells are genetically engineered T cells against specific tumor antigens.

The following slides summarize the cancers and drug classes in this presentation.⁸

Cancers and their targeted drug therapies

Cancer	Targeted Drug Therapies	
CML, Ph+	BCR:ABL TK inhibitors: Imatinib and others	
Colorectal cancer	EGFR inhibitors: cetuximab, panitumumab	VEGF inhibitor: bevacizumab
Breast cancer, HER2+	Trastuzumab, trastuzumab emtansine, pertuzumab, lapatinib	
Breast cancer, HER2–	Cyclin-dependent kinase inhibitors: Palbociclib and others	Poly(ADP-ribose) polymerase inhibitors: Olaparib and others
Lymphomas, CD20+	CD20-targeting agents: rituximab and others	
Multiple myeloma	IMiDs: Lenolidamide and others Proteasome inhibitors: bortezomib	CD36-targeting agents: darutumumab Glucocorticoid: dexamethasone
Various hematologic cancers and solid tumors	Immunotherapies CTLA-4 inhibitors: ipilimumab and others PD1 inhibitors: pembrolizumab, nivolumab PD-L1 inhibitors: atezolizumab and others	
Lymphomas, CD19+ Multiple myeloma, BCMA+	CAR-T cell therapeutics targeting CD19 or BCMA	

Inhibitors of tyrosine kinase (TK) signal transduction

BCR-ABL	TK expressed by the Philadelphia chromosome translocation (9;22)	Imatinib group are competitive inhibitors of BCR-ABL and other receptor TKs
EGFR	Epidermal growth factor receptor (structurally related TK receptors)	MAbs bind to extracellular domain and inhibit dimerization and downstream signaling
		Small molecule “tinibs” bind intracellular domain of TK receptors and inhibit autophosphorylation of the receptor, which inhibits downstream signaling
HER2	Human epidermal growth factor receptor 2 (ERBB2)	MAbs bind extracellular domain and inhibit dimerization and HER2 signaling pathway.
		Small molecule tinibs bind intracellular domain of HER2, inhibit autophosphorylation and ERBB2 signaling.
VEGF	Vascular endothelial growth factor	Biologic drugs neutralize VEGF-A, preventing activation of VEGF receptors (VEGFRs).
VDGFR	VEGF receptors	MAbs and small molecule “nibs” inhibit receptor signaling, tumor vascular permeability, enhancing blood flow and drug delivery

Inhibitors of cell cycle progression and DNA damage response

Cell cycle process that is blocked		Effects of the drugs
CDK4/6 inhibitors	Cyclin D binding with cyclin-dependent kinases CDK4 and 6 specifically enhances phosphorylation of the retinoblastoma (Rb) tumor suppressor protein, which inactivates Rb, which releases Rb’s suppression of the transcription of factors that control transition specifically from the G1 phase the S phase.	Palbociclib and the other “ciclibs” inhibit CDK4/6 which prevents progression through this cell cycle checkpoint leading to cell cycle arrest.
PARP inhibitors	Poly(ADP-ribose) polymerase (PARP) is a family of proteins that synthesize PAR, which detects and initiates immediate cellular responses to metabolic, chemical, and radiation-induced single-strand DNA breaks. PARP1 and PARP2, through multiple DNA damage response (DDR) pathways, play critical roles in DNA strand break repair that help maintain cell survival.	Olaparib and other PARP inhibitors (“paribs) in tumors with deleterious BRCA1/2 mutations or other DDR alterations prevents tumor cells from repairing DNA, making them vulnerable to apoptosis. BRCAs are tumor suppressor genes.

Drugs commonly used in the treatment of multiple myeloma

Proteasome inhibitors Bortezomib Carfilzomib Ixazomib	Inhibition of 26S proteasome: A large multi-catalytic ATP-dependent protein complex within the proteasome that degrades ubiquitinated proteins The main effect is thought to be inhibition of NF- κ B signaling pathway, the apparent major signaling pathway of multiple myeloma.
IMiDs Lenalidomide Thalidomide Pomalidomide	Immunomodulators, antiangiogenics, and antineoplastic actions by multiple mechanisms, resulting in inhibition of proinflammatory cytokine secretion, T cell stimulation. These immunomodulators induce cell cycle inhibition and cell death.
CD38 inhibitors Daratumumab Isatuximab	MAb directed against CD38, which is highly expressed on myeloma cells. It induces ADCC, complement-dependent cytotoxicity, apoptosis, and antibody-dependent phagocytosis
Dexamethasone	Glucocorticoid in multiple myeloma regimens. It induces apoptosis in multiple myeloma cells
Preferred initial standard 3-drug regimen – VRd – bortezomib, lenalidomide, dexamethasone	

Immune checkpoint inhibitors

Immune checkpoints are inhibitory signals that limit T cell activation.

CTLA-4 expressed on T cells acts as an off switch by binding CD80/86 on antigen presenting cells.

CTLA-4 inhibitor	Blocking CTLA-4 prevents the inhibitory signal and boost the T cell response against the tumors.
Ipilimumab	
Tremelimumab	

PD-1, programmed death 1, is expressed on the surface of T cells, B cells, NK cells, monocytes, and dendritic cells. PD-1 is activated by ligands PD-L1 and PD-L2.

PD-L1 is expressed on the surface of many cell types, including tumor cells.

PD-1 inhibitors

- | | |
|--|---|
| <ul style="list-style-type: none">• Pembrolizumab• Nivolumab• Cemiplimab• Dostarlimab | Blocking PD-1 unleashes the immune system response against tumor cells. |
|--|---|

PD-L1 inhibitors

- | | |
|--|--|
| <ul style="list-style-type: none">• Atezolizumab• Avelumab• Durvalumab | Blocking PD-L1 on tumor cells prevents activation of PD-1, thus, boosts the immune response against the tumor cells. |
|--|--|

Chimeric antigen receptor T cells (CAR-T cells)

Genetically engineered T cells that express artificial T cell receptors targeting specific tumor antigens. Both antigen-binding and T cell activating functions are combined in one receptor.

Anti-CD19 CAR-T cells

- Tisagenlecleucel
- Axicabtagene ciloleucel
- Brexucabtagene autoleucel
- Lisocabtagene maraleucel

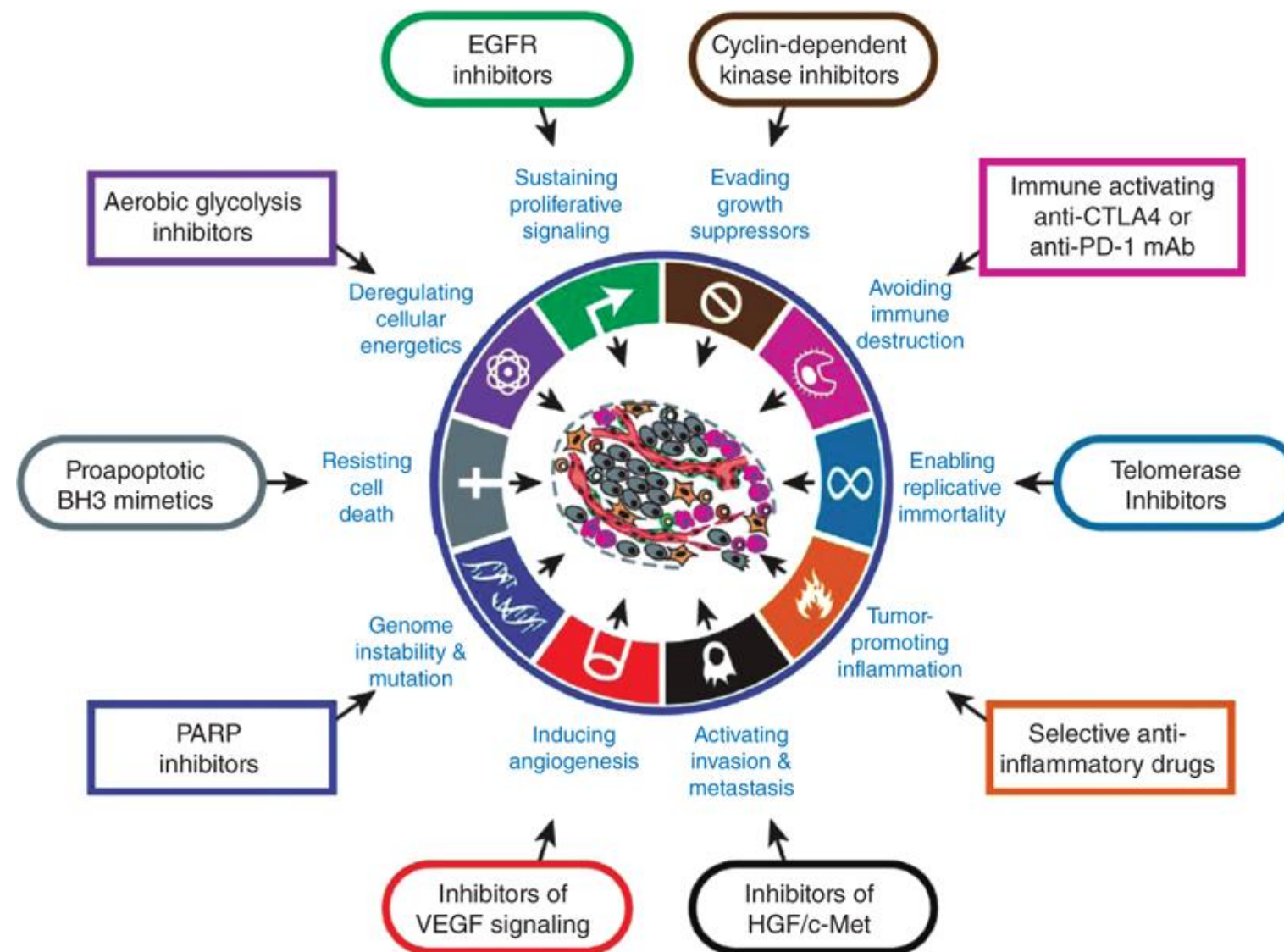
Drug-specific indications:

- B cell acute lymphoblastic leukemia
 - Diffuse large cell lymphoma
 - Follicular lymphoma
 - Mantle cell lymphoma
-

Anti-BCMA

- Idecabtagene vicleucel
 - Ciltacabtagene autoleucel
 - Multiple myeloma, dMMR
-

BCMA: B cell maturation antigen



Source: F.C. Brunicardi, D.K. Andersen, T.R. Billiar, D.L. Dunn, L.S. Kao, J.G. Hunter, J.B. Matthews, R.E. Pollock: *Schwartz's Principles of Surgery*, 11e Copyright © McGraw-Hill Education. All rights reserved.

Hallmarks of cancer and their therapeutic implications. Drugs that interfere with each of the acquired capabilities necessary for tumor growth and progression are in clinical trials and in some cases approved for clinical use in treating forms of human cancer. The drugs listed are illustrative examples. (Reproduced with permission from Hanahan D, Weinberg RA. Hallmarks of cancer: the next generation, *CCell*. 2011 Mar 4;144(5):646-674.)

Inhibitors of Receptor Tyrosine Kinase Signaling

Overview

structures, mechanisms, adverse effects

Tyrosine Kinase Signaling: NRTK and RTK

Non-receptor (soluble) tyrosine kinase (NRTK)

NRTKs are not connected to transmembrane domains.

NRTKs phosphorylate tyrosine residues on target proteins important in the regulation of a vast array of cellular functions, such as gene expression, immune response, cell survival, division.

Receptor tyrosine kinase (RTK)

RTKs are cell surface receptors for many polypeptide growth factors, cytokines, and hormones.

Ligand binding to the extracellular domain induces homo- and hetero-dimerization of the receptors, autophosphorylation on intracellular tyrosine residues, and activation of intracellular pathways.

Growth factors, like cytokines, are soluble messengers that bind cell surface receptors, which initiates transmembrane and intracellular cascade of events in signal transduction.

Tyrosine Kinase Signaling: EGFR, VEGFR and VEGF

Epidermal growth factor receptor (EGFR)

EGFR belongs to the RTK family. EGFR is activated by epidermal growth factor and transforming growth factor- α .

Mutations resulting in overexpression of EGFR lead to continuous activation of downstream signaling pathways, uncontrolled cell proliferation and survival, and tumorigenesis. EGFR overexpression is found in numerous cancers and is associated with poor prognosis.

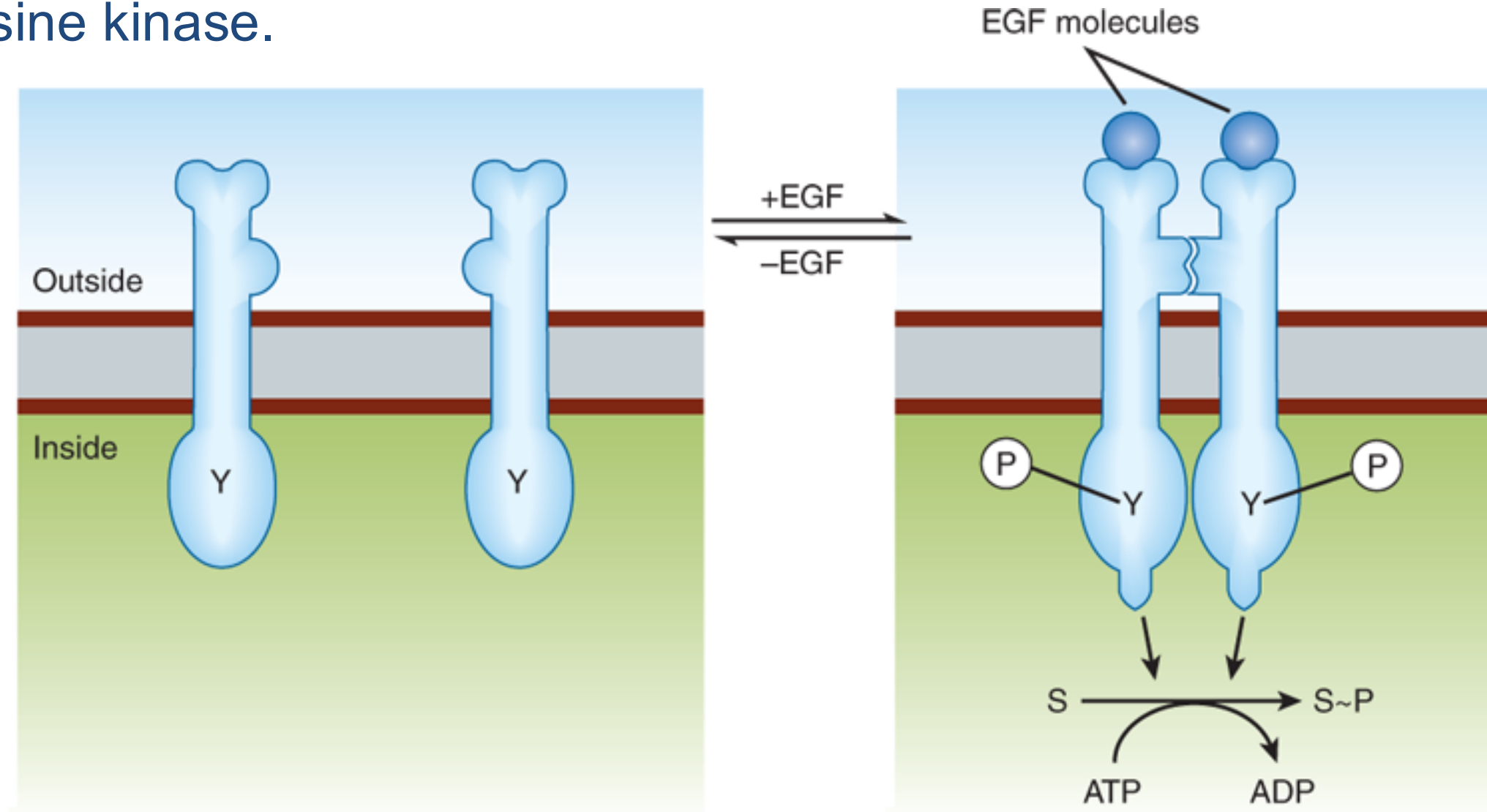
Vascular endothelial growth factor receptors (VEGFRs)

The three types of VEGFRs (-1, -2, -3) that regulate angiogenesis (new blood vessel formation) and vasculogenesis (formation of the circulatory system). Vascular endothelial growth factor-A (VEGF-A) is a member of the VEGF family and primary regulator of angiogenesis. It stimulates proliferation, migration, and survival of endothelial cells. VEGF-A is also important for vasculogenesis from endothelial precursor cells during embryonic development.

Abnormal angiogenesis occurs in cancer, age-related macular degeneration, and ischemic stroke.

Receptor tyrosine kinase (RTK) is a transmembrane receptor that binds and stimulates an intracellular protein tyrosine kinase.

1. Ligand binding (EGF example) to inactive monomeric state →
2. Active dimeric state
3. Phosphorylation (P) of specific tyrosine residues (Y) of the cytoplasmic domains
4. Activation of enzymatic catalytic activities
5. Phosphorylation of substrate proteins (S)



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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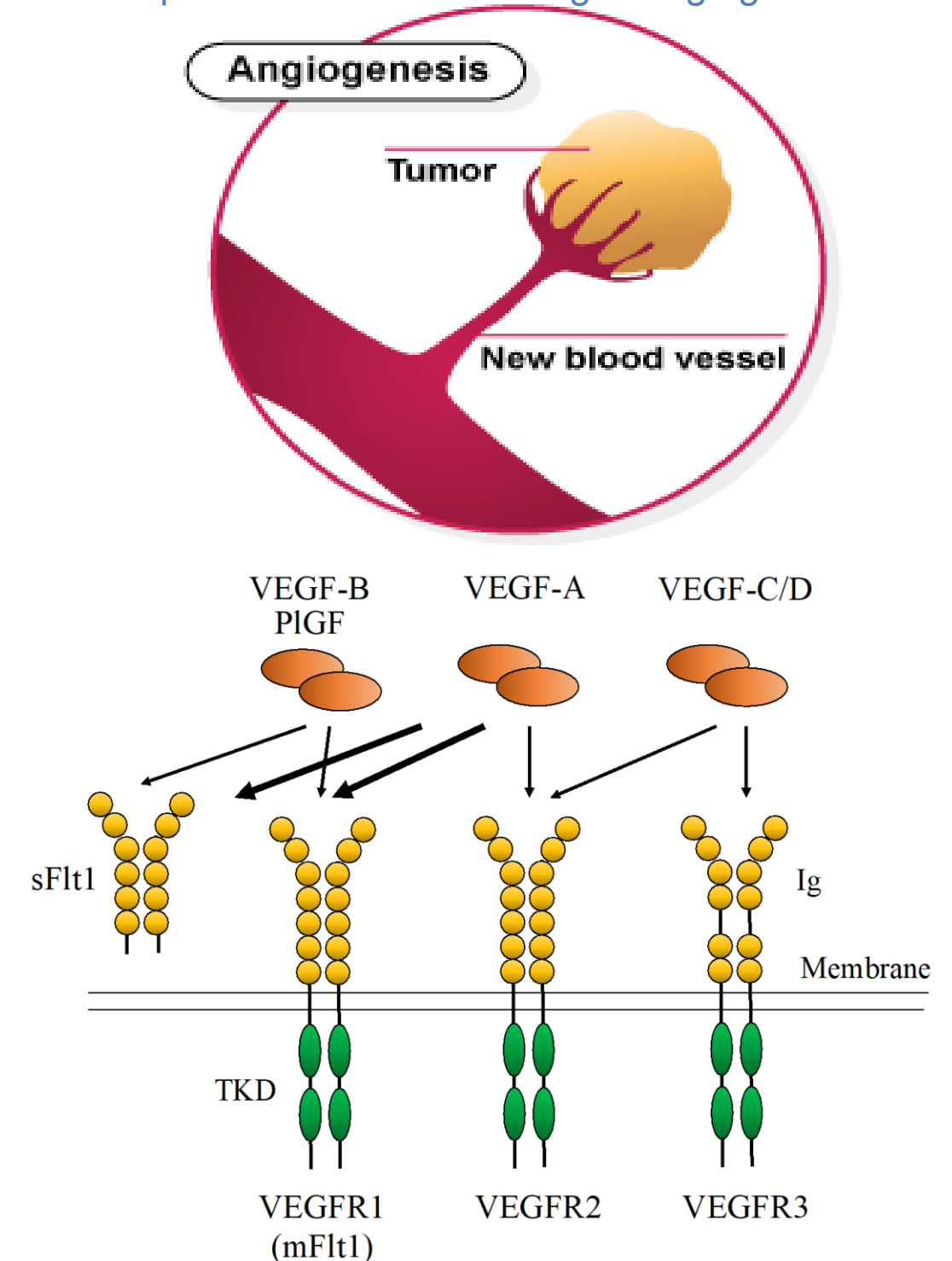
Figure adapted by LG

Angiogenesis: VEGF activates VEGFRs, which are RTKs

Tumor blood vessels have chaotic architecture and blood flow. Tumor blood flow is variable, with areas of hypoxemia and acidosis leading to the selection of variants that are resistant to hypoxemia-induced apoptosis (often due to the loss of p53 expression).

Tumor vessel walls have numerous openings, widened inter-endothelial junctions, and discontinuous or absent basement membrane.

This contributes to the high vascular permeability of these vessels and, together with lack of functional intratumoral lymphatics, causes increased interstitial pressure within the tumor, which also interferes with the delivery of therapeutics to the tumor. Tumor blood vessels lack perivascular cells such as pericytes and smooth muscle cells that normally regulate flow in response to tissue metabolic needs.



Mechanisms: Receptor Tyrosine Kinase Inhibitors (RTKIs)

Small molecules	Passively diffuse into cytosol Nomenclature: All have the suffix “tinibs” or “nibs”
Mechanism	Block the intracellular ATP binding sites of the tyrosine kinase domain, preventing autophosphorylation and activation of signaling pathways
Oral CYP3A4 substrates	Oral route and once or twice daily dosing is convenient for the patient. High potential for drug-drug interactions. Many are also substrates of P-gp.
Adverse effects	Mechanism related: Inhibition of growth pathways in normal cells • Nausea, vomiting, diarrhea • Rash (acneiform) / desquamation (skin peeling) • Liver injury • Myocardial dysfunction / heart failure • Bone marrow dysfunction: anemia, neutropenia, thrombocytopenia • Susceptibility to infection • Hypertension • Periorbital edema • and more

Mechanisms: Antibody Inhibitors of Tyrosine Kinase Receptors

Proteins	Monoclonal antibodies nomenclature suffix “mabs”
Mechanism	Antibodies bind the bind the extracellular domain at specific epitopes of specific receptor types, which blocks the natural ligand from binding and prevents receptor dimerization and activation. (Induction of anti-drug antibodies reduces efficacy.)
Injection long half-life	IV continuous infusion or subQ injection Coformulation with hyaluronidase permits subcutaneous injection by catalyzing degradation of hyaluronic acid, making tissues more permeable Half-life days to weeks (drug-specific) Metabolized mainly by proteolysis in the liver and reticuloendothelial system
Adverse effects	Mechanism related: Inhibition of growth pathways in normal cells • Class adverse effects (previous slide) • Infusion reactions, systemic and can be severe; can occur during or shortly after infusion; premedication with antihistamine and glucocorticoid can mitigate this adverse effect • Injection site reactions, local • Hypersensitivity reactions

Mechanisms: Inhibitors of VEGF and VEGFR

Structures	Monoclonal antibodies Small molecule multitargeted drugs, oral
Mechanisms	1- Antibody binds soluble extracellular VEGF-A (a growth factor) preventing it from activating its receptors 2- Antibody binds VEGFR inhibiting dimerization and activation of signaling pathways Small molecule multitargeted drugs block various signaling pathways, including the angiogenesis pathways.
Adverse effects	Mechanism related: Inhibition of angiogenesis • Hypertension • Hemorrhage • Arterial thromboembolism • Impaired wound healing • Wound dehiscence (separation of previously closed wound) • Infusion reactions

BCR-ABL Kinase Inhibitors

For the treatment of chronic myeloid leukemia (CML)

*Imatinib	• Dasatinib	• Nilotinib
*Ponatinib	• Bosutinib	• Asciminib

The second-generation inhibitors (all those developed after imatinib) are more potent than imatinib and are effective against some resistance-conferring mutations.

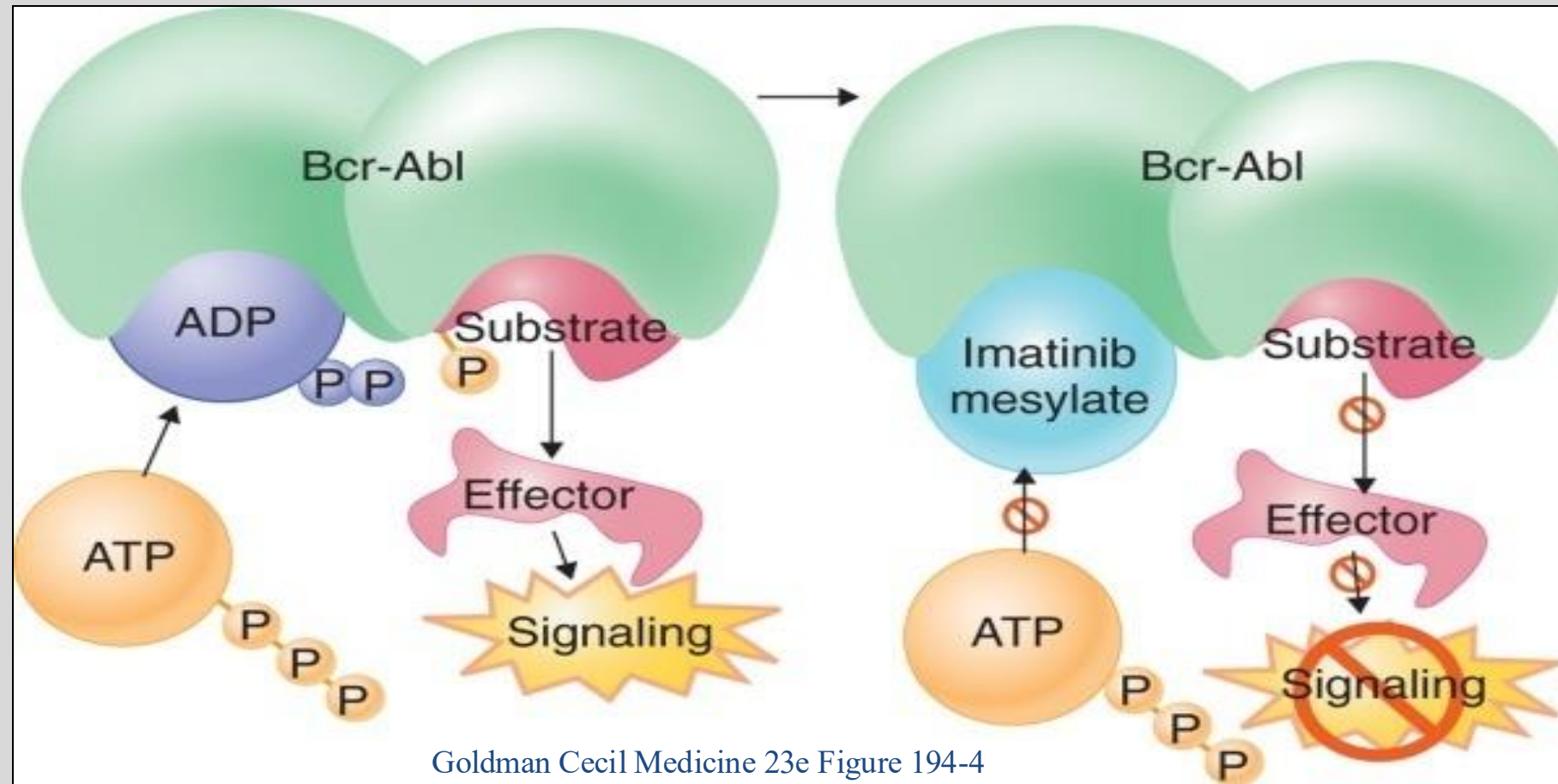
Tumor cells harboring the **T315I** mutation is susceptible only to **ponatinib** and **asciminib** due to different mechanisms of action. (T315I: threonine to isoleucine point mutation)

BCR::ABL1 fusion oncogene – the Philadelphia chromosome – is a clonal cytogenetic abnormality resulting from a translocation between chromosomes 9 and 22. It codes a constitutively active cytoplasmic (NRTK) tyrosine kinase, BCR::ABL1, that drives unregulated cell growth.

Class Properties: Typical of all “tinibs”

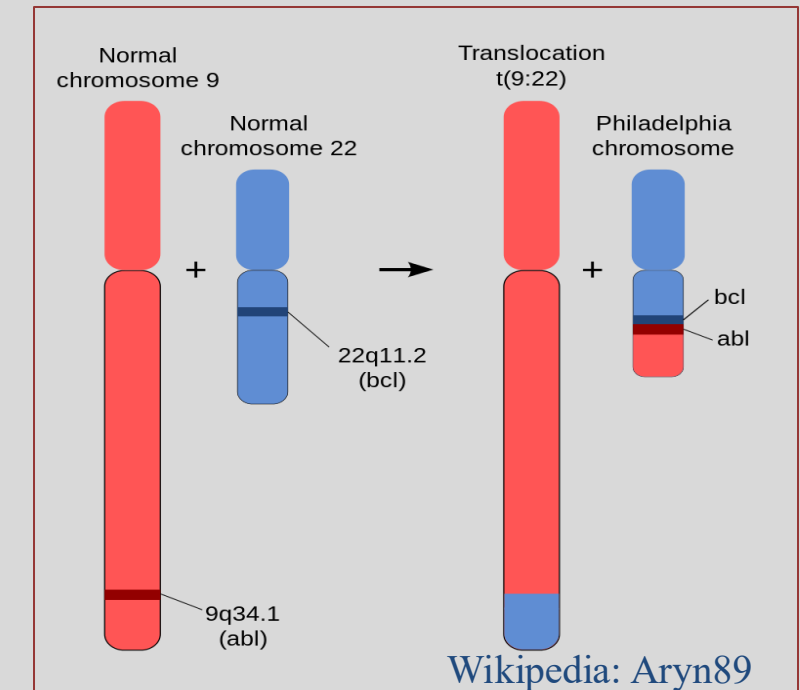
A	Oral Protein binding >90% CYP3A4, P-glycoprotein substrate excreted in feces
MOA	Binds TK and prevents ATP from binding and activating the kinase (see next slide)
MOR	Point mutations prevent tight binding of the drug and lock the enzyme in its open configuration, allowing the substrate access to the site. Point mutations can arise in 3 separate segments of the kinase domain (main mechanism) T315I in ABL1 residue: Only ponatinib and asciminib are effective.
Uses	Imatinib: CML, Ph+, initial chronic, accelerated or blast phase, GIST and other c-kit pathways, PGDFR-driven cancers (see following chart)
AEs	Typical of tinibs class: nausea/vomiting/diarrhea; bone marrow suppression; hepatotoxicity • Imatinib-induced periorbital edema occurs frequently. • Skin rashes occur relatively frequently. Mild rashes are managed with topical steroids. Severe rashes require dose reduction or temporary discontinuation of the drug • Drug interactions, high potential, CYP3A4 and P-glycoprotein related

Inhibition of the tyrosine kinase domain of the BCR-ABL oncoprotein



Goldman Cecil Medicine 23e Figure 194-4

9:22 translocation



Wikipedia: Aryn89

BCR::ABL1 inhibitors occupy the ATP binding pocket → prevent substrate phosphorylation and downstream activation of signals

- Imatinib and the other BCR::ABL1 inhibitors also inhibit various tyrosine kinases, such as platelet-derived growth factor receptor (PDGFR) and c-kit (drug-specific actions).
- Ponatinib has multiple contact points → an individual mutation has less effect on overall binding.
- Asciminib is an allosteric inhibitor → holds BCR::ABL1 in an inactive conformation.

BCR-ABL kinase inhibitors Therapeutic Uses

Goals, patients with chronic phase CML: To achieve clinical remission, maintain long-term disease control, avoid disease progression to accelerated phase or blast crisis, and optimizing the quality of life by limiting drug-related toxicities

Drug	CML, Ph+	ALL, Ph+ note correction	CML, T315I mutation	Other
Imatinib	✓*	✓		GIST (c-Kit), other c-Kit tumors PDGFR-driven cancers
Dasatinib	✓*	✓		GIST
Nilotinib	✓*	✓		GIST
Bosutinib	✓*			
Ponatinib	✓*	✓	✓*	
Asciminib	✓		✓	

*initial chronic, accelerated or blast phase

GIST: gastrointestinal stromal tumor; PDGFR: platelet-derived growth factor receptor



Open i beta: An open access biomedical search engine <http://openi.nlm.nih.gov/index.php>
Imatinib-induced rash and periorbital edema

Check your knowledge:

You might want to pause the video to think about the answers.

- What type of protein is expressed by the *BCR::ABL1* driver mutation?
- What is the importance of the T315I point mutation in the ABL1 contact residue?
- What drugs are active against CML harboring this mutation?
- What are some class toxicities of the tinibs group?
- What are the serious and potentially life-threatening toxicities of ponatinib?

Biologic Inhibitors

For the treatment of metastatic colorectal cancer

Goals of therapy:

To prolong overall survival and maintain quality of life for as long as possible

EGFR Inhibitors

*Cetuximab

Panitumumab

VEGF Inhibitor

*Bevacizumab

Adjuvant therapy with chemotherapy such as FOLFOX

Pharmacokinetics of MAbs: parenteral, endocytosis and proteolysis

Cetuximab and Panitumumab

Competitive inhibitors of EGFR, HER1, ErB-1



block dimerization which prevents
phosphorylation and activation of
receptor-associated kinases



Inhibits cell growth
Induces apoptosis
Decreases matrix metalloproteinase
Decreases VEGF

Cetuximab also induces ADCC.

Cetuximab and panitumumab are NOT effective in tumors expressing mutations of *KRAS* oncogene (occurs in 40-50% of colorectal tumors) → tumors should be tested for *KRAS* mutations.

Bevacizumab

Binds and neutralizes VEGF-A
(takes VEGF-A out of action)



prevents VEGF-A binding to
endothelial VEGF receptors



prevents receptor signaling that normally
increases vascular permeability
and formation of new blood vessels

Therapeutic Uses

Cetuximab:

- Colorectal cancer, metastatic, EGFR-expressing, KRAS wild-type
1st-line with FOLFOX or FOLFIRI
- Head and neck cancer, squamous cell with cisplatin-based
- Penile cancer, squamous cell
- Skin cancer, squamous cell,

Panitumumab:

- Colorectal cancer, metastatic, EGFR-expressing, KRAS wild-type with FOLFOX or FOLFIRI or other chemo
- Skin cancer, squamous cell, advanced

Adverse effects are typical class toxicities related to inhibition of EGFR signaling and certain drug-specific effects of certain combination regimens

Bevacizumab:

- Colorectal cancer, metastatic
- Cervical cancer, metastatic
- Glioblastoma, recurrent
- Hepatocellular carcinoma, metastatic
- Non-small cell lung cancer, nonsquamous
- Ovarian epithelial, fallopian tube, primary peritoneal cancer
- Renal cell carcinoma, metastatic

Adverse effects are related to impairment of angiogenesis including severe hemorrhage, poor wound healing / wound dehiscence, hypertension, arterial occlusive events: stroke, MI, GI perforation

Check your knowledge:

You might want to pause the video to think about the answers.

- Cetuximab and panitumumab are ineffective in tumors with what mutation?
- What are the target and disease treated by both cetuximab and panitumumab?
- What is the mechanism of bevacizumab?
- What cytotoxic chemotherapies are included in the combination regimen?
- What are some serious and potentially life-threatening toxicities of these MAbs?

HER2 Inhibitors

For the treatment of HER-2-positive breast cancer

Monoclonal Antibodies	MAb-Cytotoxic Drug Conjugates	Small Molecule Targeted Drugs
*Trastuzumab *Pertuzumab Margetuximab	*Ado-trastuzumab emtansine Fam-trastuzumab deruxtecan	*Lapatinib Neratinib

Breast cancer is categorized as 1) those expressing estrogen receptor (ER), 2) those expressing HER2, with or without ER expression, and 3) triple-negative – those that do not express ER, HER2 or progesterone receptor.

The *ERBB2* gene expresses HER2 RTK. *HER2* oncogene is present in about 15% of primary invasive breast cancers and is associated with lower response rates to hormonal therapies and to most cytotoxic drugs, with the exception of anthracyclines.

Trastuzumab and Pertuzumab Mechanism of Action

The pertuzumab-trastuzumab combination provides more complete inhibition of HER2 signaling.

Trastuzumab

Binds HER2 domain IV



- 1-Prevents ligand-independent dimerization,
- 2-Induces receptor-MAb internalization and degradation, and
- 3-Induces ADCC

Pertuzumab

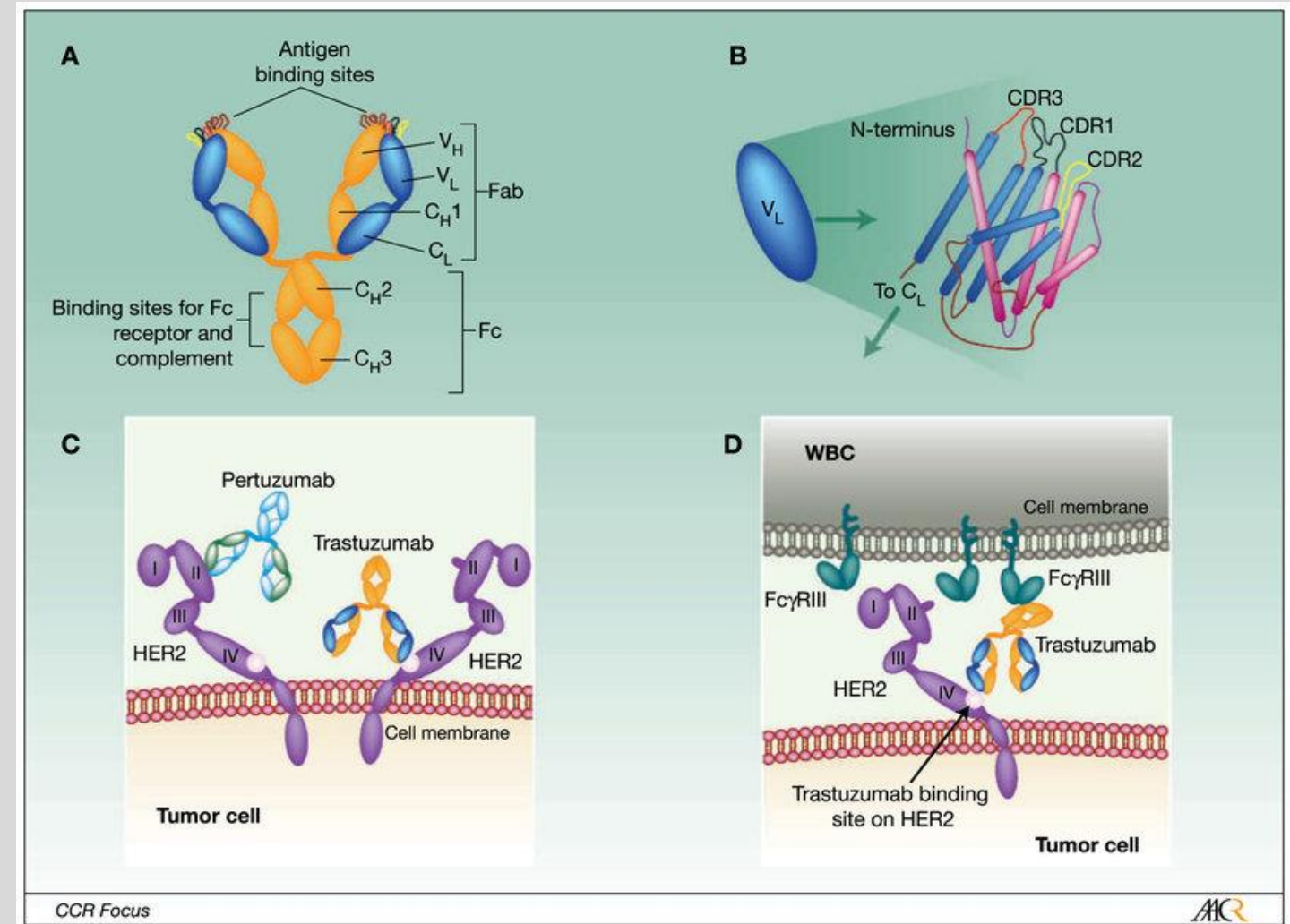
Binds HER2 domain II



- 1-Inhibits ligand-dependent heterodimerization with other HERs
- ↓
- Inhibits signaling, cell growth, and survival and
- 2-Induces ADCC

ADCC: Antibody-dependent cellular cytotoxicity

Margetuximab: Engineered to have greater ADCC and activation of NK cells compared to trastuzumab.



CCR Focus

AKR

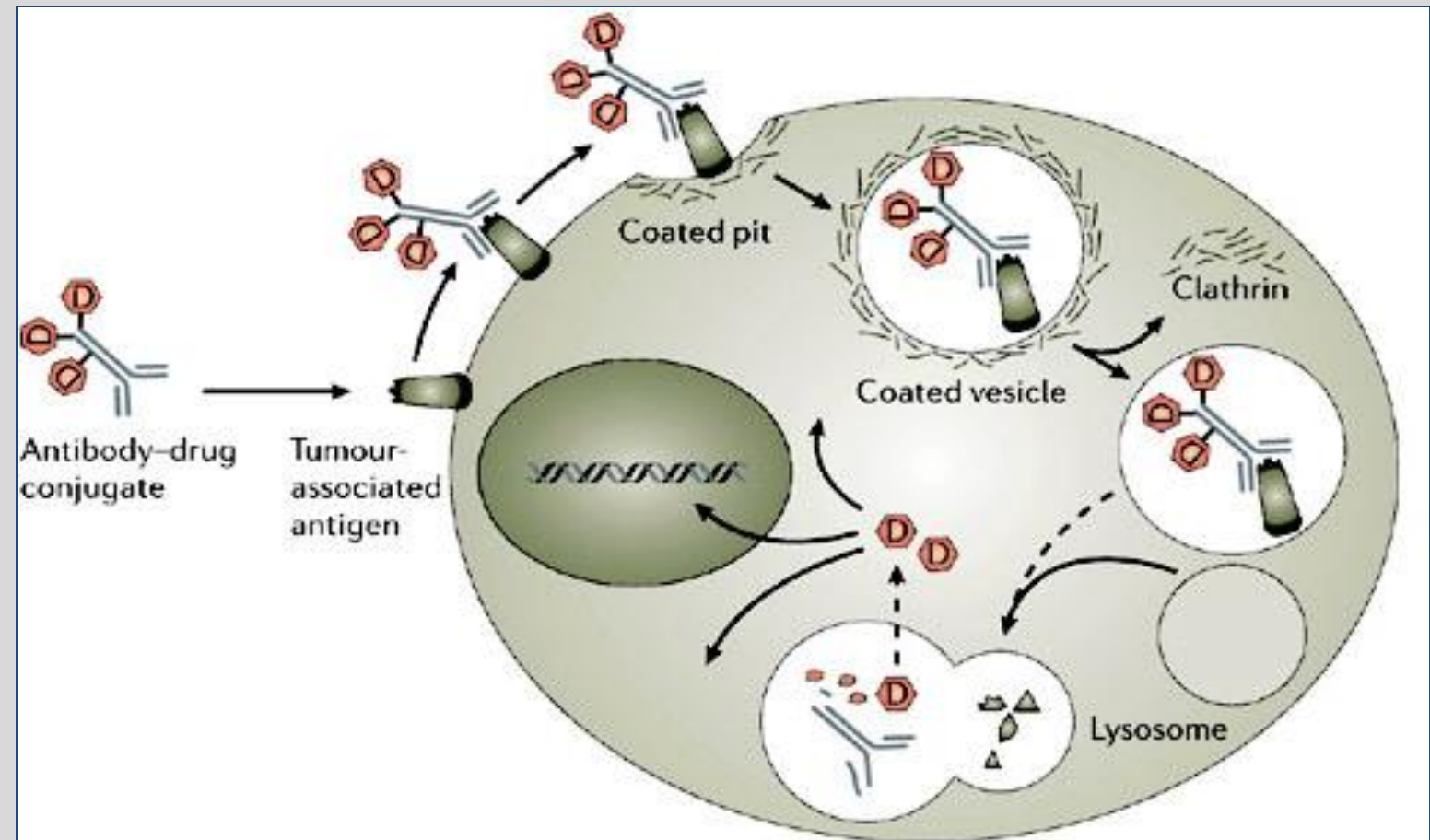
Trastuzumab schematic structure Pohlman P, Mayer IA, Mernaugh R. Resistance to Trastuzumab in Breast Cancer. Clinical Cancer Research, Dec 2009, 15(24):7479-7491
https://www.researchgate.net/publication/40686569_Resistance_to_Trastuzumab_in_Breast_Cancer

Ado-Trastuzumab Emtansine (DM1) Antibody-Drug Conjugate

1. Inhibits HER2 signaling, and...
2. Internalization of receptor delivers the chemotherapeutic DM1 directly inside HER2-positive cancer cells →
3. Inhibits microtubule assembly by binding to tubulin →
4. Results in cell cycle arrest and apoptosis.

Fam-trastuzumab-deruxtecan:

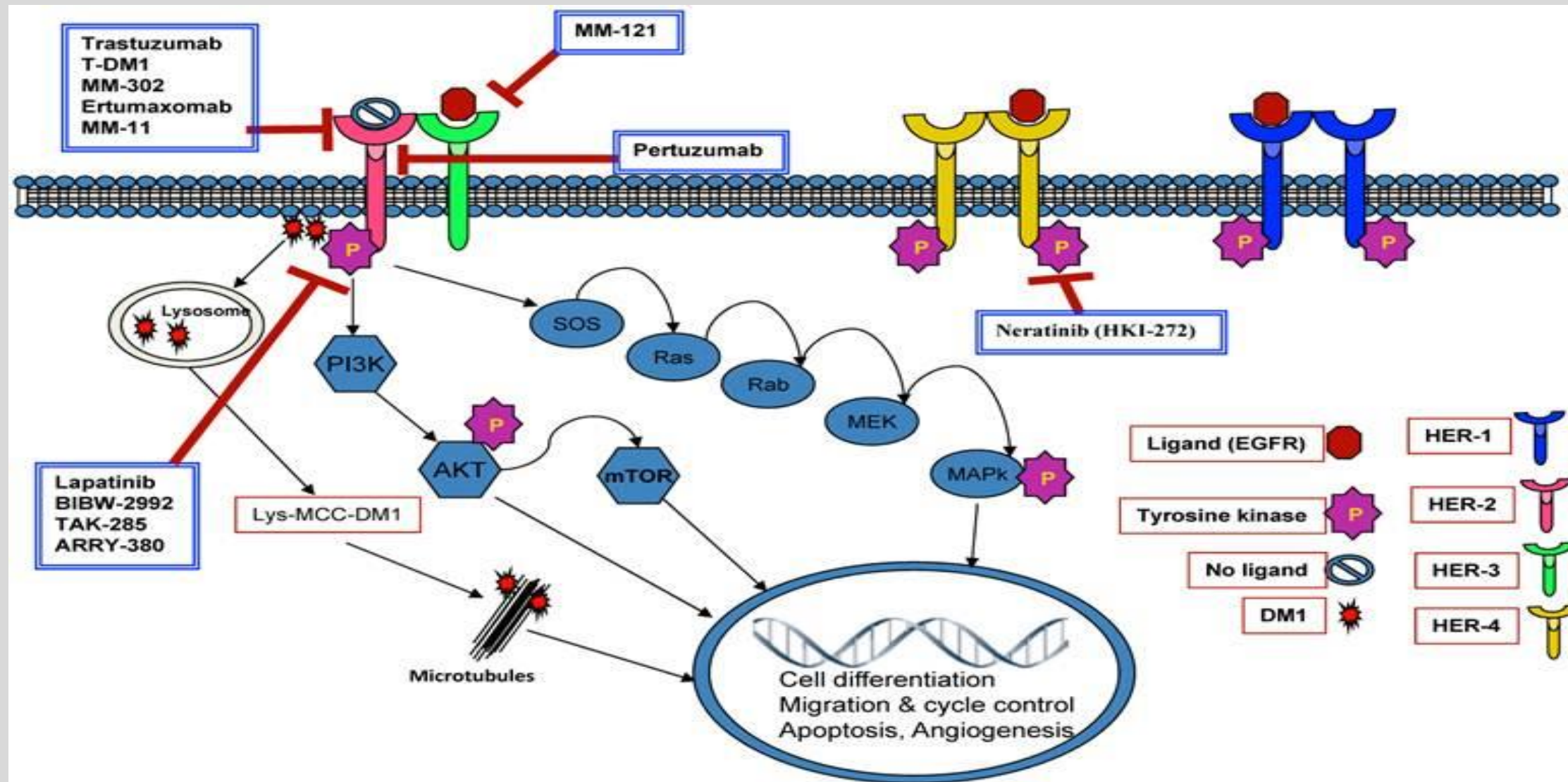
Cytotoxin is a topoisomerase I inhibitor.



<http://www.blogoup.com/blog/2011/2/7/antibody-drug-conjugates.html>

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Nature Reviews | Drug Discovery

Therapies Targeting HER2 Signaling



Resistance:

Different mechanisms
probably coexist:
Target alterations,
PI3K-AKT pathway
alterations,
HER2 overexpression

Treatment of HER2-positive tumors In combination with cytotoxic chemotherapy

Trastuzumab / Trastuzumab-hyaluronidase

Neo-adjuvant and adjuvant therapy

- Breast cancer

First-line adjuvant treatment regimen:

Trastuzumab, taxane, and/or doxorubicin,
cyclophosphamide, $\pm$ pertuzumab

- Gastric cancer
- Colorectal cancer
- Endometrial cancer

Pertuzumab

Neo-adjuvant and adjuvant therapy

- Breast cancer, early and metastatic

Pertuzumab-trastuzumab-hyaluronidase

- SubQ in patients who have not received prior anti-HER2 or chemotherapy for metastatic disease

Hyaluronidase increases permeability of subcutaneous tissue.

Ado-Trastuzumab Emtansine

- Breast cancer, early and metastatic
- in patients previously treated with trastuzumab and taxane

Trastuzumab / Pertuzumab Toxicities and Drug Interactions

Cardiac failure the most serious toxicity Clinical and subclinical	Decreased left ventricular ejection fraction and congestive heart failure Highest risk with anthracycline chemotherapy regimen Underlying cardiac disease should be ruled out prior to starting therapy; clinical follow-up thereafter.
Infusion reactions	Fever, chills, nausea, dyspnea, rash; typical of monoclonal antibody therapy
Pulmonary toxicity	Pulmonary infiltrates / edema / Interstitial pneumonitis / ARDS
Various others	GI problems, peripheral sensory neuropathy, joint/muscle pain, elevated serum transaminases, and more
Pregnancy	Embryo-fetal death
Drug interactions	Doxorubicin, cyclophosphamide ↑ risk of cardiac toxicity
Ado-trastuzumab	Hepatotoxicity w. liver failure/death, thrombocytopenia, sensory and motor peripheral neuropathy, CYP3A4 drug interactions

Lapatinib: Small Molecule TKI targeting HER2

PK	Oral protein binding >99% hepatic CYP3A4 feces t _{1/2} ~24 h
MOA	Reversibly binds EGFR (ErB1) and HER2 (ErB2) → blocks autophosphorylation of tyrosine residues → blocks activation of second messengers, Erk1/2 and Akt
Resistance	HER2 TK domain mutations, alternate receptors, alternate intracellular pathways
Uses	Breast cancer, advanced or metastatic (combination regimens) • After prior anthracycline, taxane, trastuzumab therapy • With progression on prior trastuzumab therapy • With previously untreated brain metastases • Salvage therapy Colorectal cancer, metastatic, with progression on conventional chemotx
Toxicities	Diarrhea, rash, cardiotoxicity ↓LVEF, QT prolongation, hepatotoxicity, pulmonary toxicity
Interactions	CYP3A4-related

Check your knowledge:

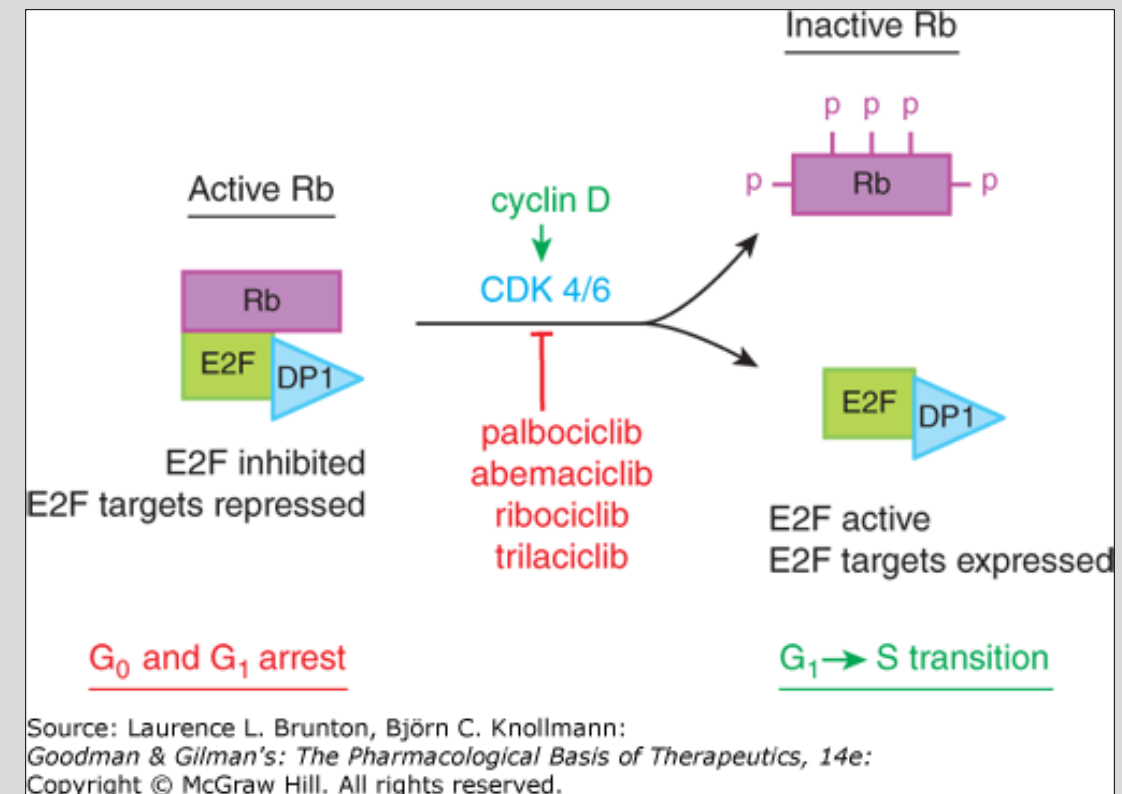
You might want to pause the video to think about the answers.

- Mechanistically, how do trastuzumab and pertuzumab work together?
- Mechanistically, how does lapatinib interfere with HER2 signaling?
- What is the mechanistic concept of antibody-drug conjugates?
- What cytotoxic chemotherapies are included in the combination regimen?
- What are some serious and potentially life-threatening toxicities of trastuzumab?

Cyclin-Dependent Kinase (CDK) 4/6 Inhibitors (“Cyclibs”)

For the treatment of
hormone receptor-positive,
HER-2-negative breast cancer

- *Palbociclib
- Ribociclib
- Abemaciclib



Cyclin D binding with CDK4 and 6 specifically enhances phosphorylation of the retinoblastoma tumor suppressor protein (Rb), inactivating Rb, which releases Rb's suppression of the E2F-DP1 transcription factor that regulates transition specifically from the G1 phase the S phase. The cell is allowed to proceed through the cell cycle and divide.

Cyclin D-CDK4/6 pathway appears to be hyperactivated in human hormone receptor-positive breast cancers.

Ciclibs pharmacology

PK	Oral protein binding 85-95%; Pgp hepatic CYP3A4 feces t _{1/2} ~24 h
MOA	Reversible inhibitors of CDK4 and CDK6 → prevents progression from G1 to S cell cycle phase, leading to cell cycle arrest.
Uses	Breast cancer, hormone receptor-positive, HER2-negative, advanced or metastatic • Preferred initial therapy: CDK4/6 inhibitor plus aromatase inhibitor • As effective and less toxic than chemotherapy regimen Aromatase converts androgens to estrogens. Blocking the enzyme results in estrogen depletion.
Toxicities (common)	Neutropenia, leukopenia, anemia, thrombocytopenia; skin reactions, pulmonary toxicity, infection, stomatitis, fatigue, diarrhea, nausea, headache Ribociclib: QT interval prolongation; liver function abnormalities
Interactions	CYP3A4 and P-glycoprotein-related

*PARP INHIBITORS “Paribs”

For the oral treatment of BRCA-mutated advanced cancers

Cancers, BCRA mutated	*Olaparib	Niraparib	Rucaparib	Talazoparib
Breast cancer, metastatic, HER2-negative	√			√
Fallopian tube cancer	√	√	√	
Ovarian cancer	√	√	√	
Pancreatic cancer	√			
Peritoneal cancer, primary	√	√	√	
Prostate cancer, castration-resistant	√		√	

Toxicities:

- Common: Nausea, thrombocytopenia, anemia, leukopenia, dyspnea, diarrhea, CYP-mediated drug interactions
- Risk for myelodysplastic syndrome and AML (0.3% of patients)

Poly(ADP-ribose) (PAR) is synthesized from NAD⁺ by PAR polymerases (PARPs) located in the cell nucleus. PAR detects and initiates immediate cellular responses to metabolic, chemical, or radiation-induced single-strand DNA breaks.

PARP1 and PARP2, through multiple DNA damage response (DDR) pathways, play critical roles in DNA strand break repair that help maintain cell survival.

The use of PARP inhibitors in tumors with deleterious BRCA1/2 mutations or other DDR alterations prevents tumor cells from repairing DNA, making them vulnerable to apoptosis.

Anti-CD20 Agents

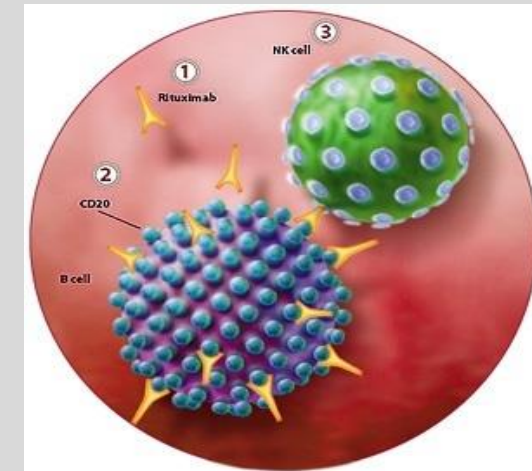
For the treatment of CD20-positive B cell malignancies

*Rituximab (chimeric IgG1MAb)

Ofatumumab Obinutuzumab Ocrelizumab for multiple sclerosis

^{90}Y Ibritumomab tiuxetan

radioimmunoconjugate for the treatment of relapsed indolent lymphomas

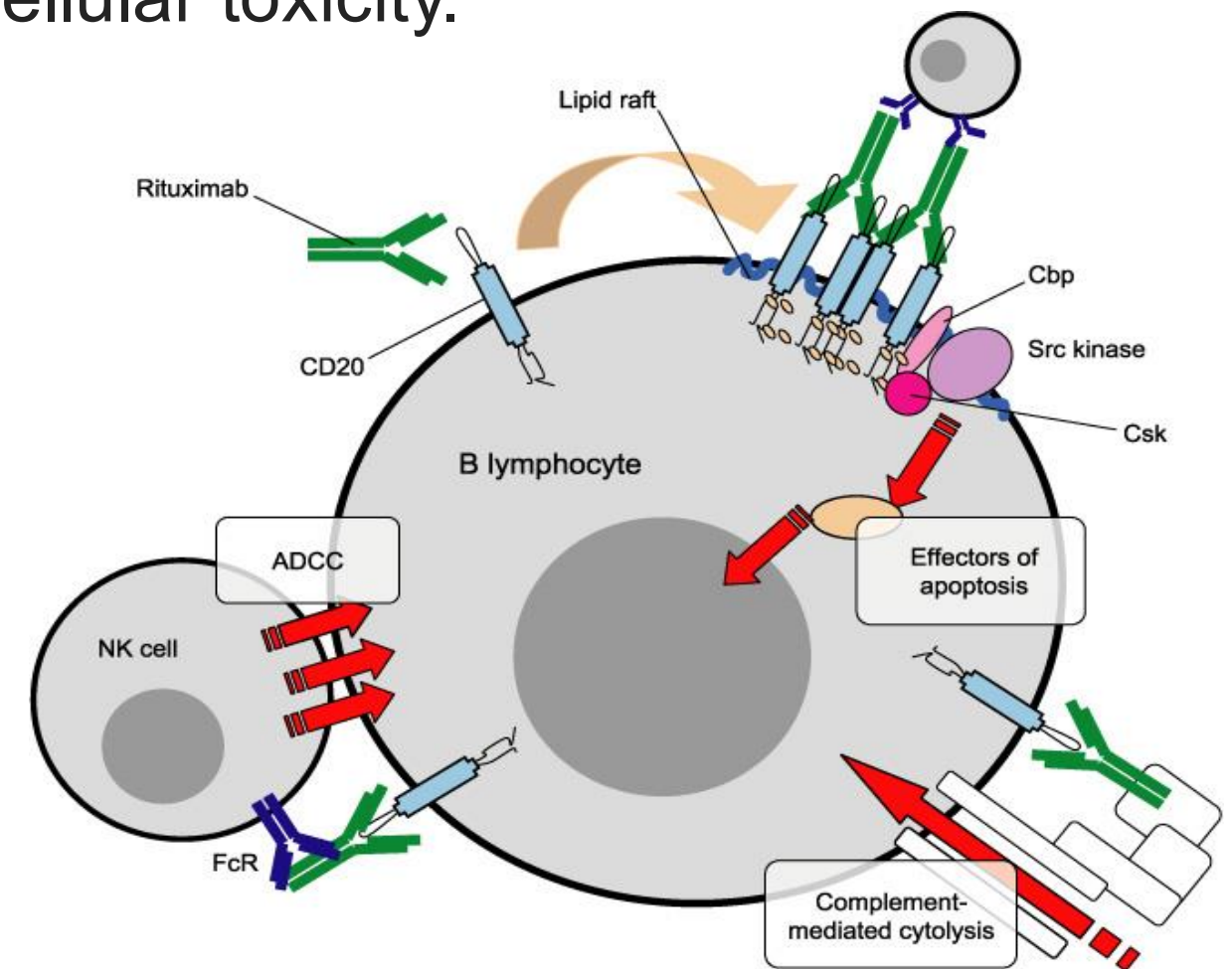
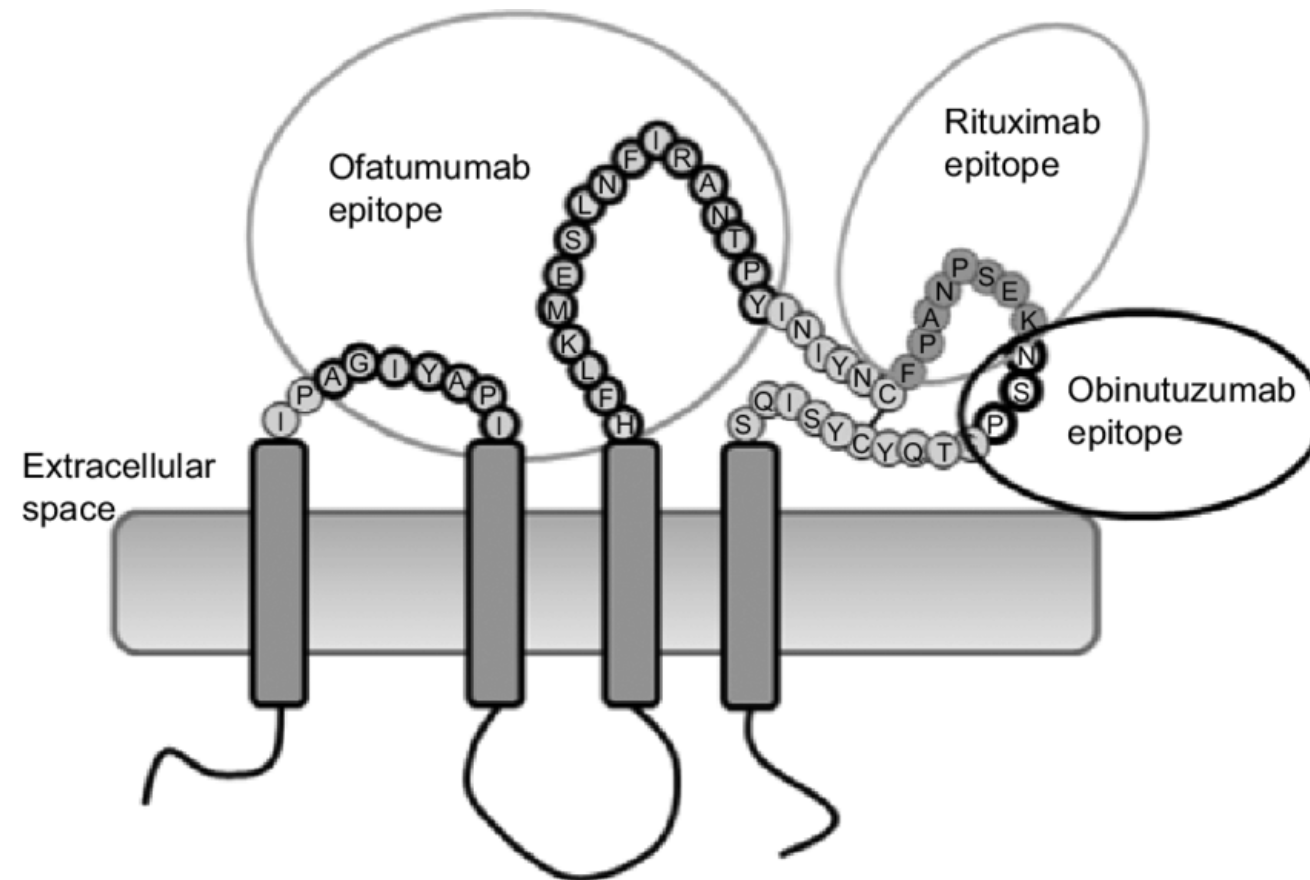


- CD20 is a transmembrane protein expressed on 95% of B cells in peripheral blood and secondary lymphoid organs. CD20 is absent on terminally differentiated plasma cells.
- CD20 is a useful biomarker, as it is expressed on many B cell malignancies, such as chronic lymphocytic leukemia, diffuse large B cell lymphoma, mantle cell lymphoma, and follicular lymphoma.
- B lymphocytes perpetuate the inflammatory process in autoimmune disorders. Anti-CD20 therapies are useful in the management of several of these diseases

Rituximab binds to extracellular loop of CD20 → cellular cytotoxicity
Rituximab induces depletion of peripheral B cells.

1. Activates complement-dependent B-cell cytotoxicity

2. Binding Fc receptors mediates cell killing through an antibody-dependent cellular toxicity.



https://www.researchgate.net/figure/Structure-of-CD20-and-epitope-targets-of-ofatumumab-rituximab-and-obinutuzumab-GA101_fig1_281779437

Klein C, Lammens A, Schafer W, et al. Epitope interactions of monoclonal antibodies targeting CD20 and their relationship to functional properties. MAbs. 2013;5(1):22-33. 15

<http://www.inviaspeciale.com/wp-content/uploads/2009/03/rituximab.jpeg>

Rituximab

PK: I.V.; rituximab-hyaluronidase → subQ admin

ANTICANCER USES NHL, CLL, Mantle cell lymphoma, ALL–B cell precursor

Example of standard therapy of diffuse large B cell lymphoma:

R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisone)

IMMUNE DISORDERS

Steroid-refractory chronic GVHD	Post-transplant lymphoproliferative disorder (PTLD)
Rheumatoid arthritis	Autoimmune hemolytic anemia (AIHA) in children
Refractory pemphigus vulgaris	Chronic immune thrombocytopenic purpura (ITP)
Refractory lupus nephritis	and more

ADVERSE EFFECTS

Possible severe infusion reactions (first-dose)	Tumor lysis syndrome
Opportunistic infection	Severe skin reactions
	Reactivation of latent HBV infection; PML

Practice point: All patients should be screened for HBV infection. In high-risk populations, screen for latent infections, including tuberculosis, hepatitis C virus, and HIV.

Check your knowledge:

You might want to pause the video to think about the answers.

- Rituximab targets what antigen on what types of immune cells?
- What is rituximab's mechanism of tumor and autoimmune cell eradication?
- Patients should be screened for what infections prior to beginning rituximab tx?
- What is the purpose of administering rituximab in conjunction with hyaluronidase?
- How do CDK4/6 inhibitors and PARP inhibitors cause tumor cell death?

Multiple Myeloma Pharmacology

Immunomodulatory Imide Drugs (IMiDs)

Proteasome inhibitors

CD38 inhibitors

Dexamethasone

Proteasome Inhibitors, IMiDs, Anti-CD38

For the treatment of multiple myeloma

Immunomodulatory Imide Drugs (IMiDs)	Proteasome inhibitors	Anti-CD38 MAb
*Lenalidomide (Revlimid)	*Bortezomib (Velcade)	*Darutumumab
Thalidomide	Carfilzomib	Isatuximab
Pomalidamide	Ixazomib	

Additional
targeted agents
have been
developed for
the treatment of
relapsed or
refractory
multiple
myeloma.

Preferred induction therapy based on risk stratification:

Standard risk: VRd or DRd (d=dexamethasone)

High-risk, autologous HCT eligible: DVRd, VRd, or DRd

Multiple myeloma is a plasma cell neoplasm characterized by clonal plasma cells that produce a monoclonal immunoglobulin. These plasma cells proliferate in bone marrow and can cause osteolytic lesions, osteopenia, and/or pathologic fractures. Hypercalcemia, kidney impairment, anemia, and infections are additional disease-related complications.

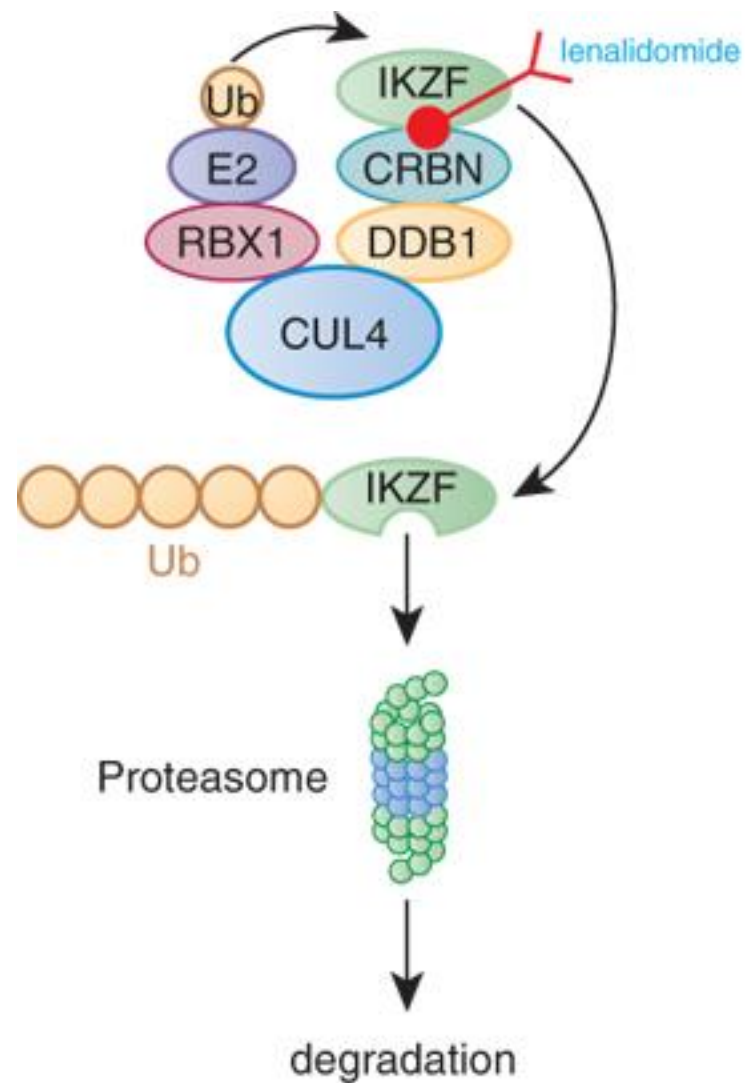
The goal of treatment is to alleviate symptoms, reverse cytopenias, decrease end-organ damage, achieve sustained response, improve quality of life, and achieve sustained overall survival. The drugs presented here are most frequently used.

IMiDs Pharmacokinetics

*Lenalidomide

PK	Oral; 1x daily for 21d of 28-day cycle
MOA	Binds cereblon, a protein crucial in tagging proteins for degradation via the ubiquitin-proteasome pathway. IMiDs prevent release of NF κ B, which reduces inflammation and promotes apoptosis in cancer cells.
TU	Lenalidomide is preferred IMiD for treatment of multiple myeloma, myelodysplastic syndrome, and CLL. It is more potent and has a broader range of therapeutic uses than the other IMiDs.
AEs	• Heme: Neutropenia, Thrombocytopenia • Thromboembolic events: VTE, DVT, PE, acute MI, stroke • Peripheral neuropathy, sensorimotor • Hepatotoxicity: Abnormal LFTs, hyperbilirubinemia • Hypersensitivity reactions: Immediate: anaphylaxis, angioedema (immediate) and SJS/TEN/DRESS (delayed) PREGNANCY Contraindicated: fetotoxic and teratogenic – causes phocomelia

Thalidomide and pomalidamide have similar PK properties (pomalidamide: hepatic metabolism via CYP1A2 / 3A4), the same mechanism of action and adverse effects.



Source: Laurence L. Brunton, Björn C. Knollmann:
Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e:
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Lenalidomide (LEN) Mechanism of Action

LEN binds cereblon (CRBN)



forms LEN-CRBN complex



recruits substrates to CRBN for ubiquitination
 by E3 ubiquitin-protein ligase



proteasome degradation of transcription factors IKZF



death of multiple myeloma cells

Numerous IMiD-dependent CRBN substrates have been identified.

Definitions are on the next slide.

Recruitment of target substrates by lenalidomide. Lenalidomide functions as a molecular “glue” for the recruitment of proteins to the cereblon (CRBN) receptor component of the cullin–RING (Really Interesting New Gene) E3 ubiquitin ligase complex 4 (CRL4) comprising cullin 4 (CUL4), DNA damage-binding protein 1 (DDB1), and E3 ubiquitin–protein ligase RBX1. This recruitment results in protein ubiquitylation and degradation. Due to the structural differences, thalidomide, and its congeners, interact with and lead to the depletion of overlapping but distinct sets of proteins that include IKZF1/3 (Ikaros family zinc finger protein) and CK1α (casein kinase I isoform-α) (Jan et al., 2021).

Definitions

Ubiquitin: 76 amino acid protein

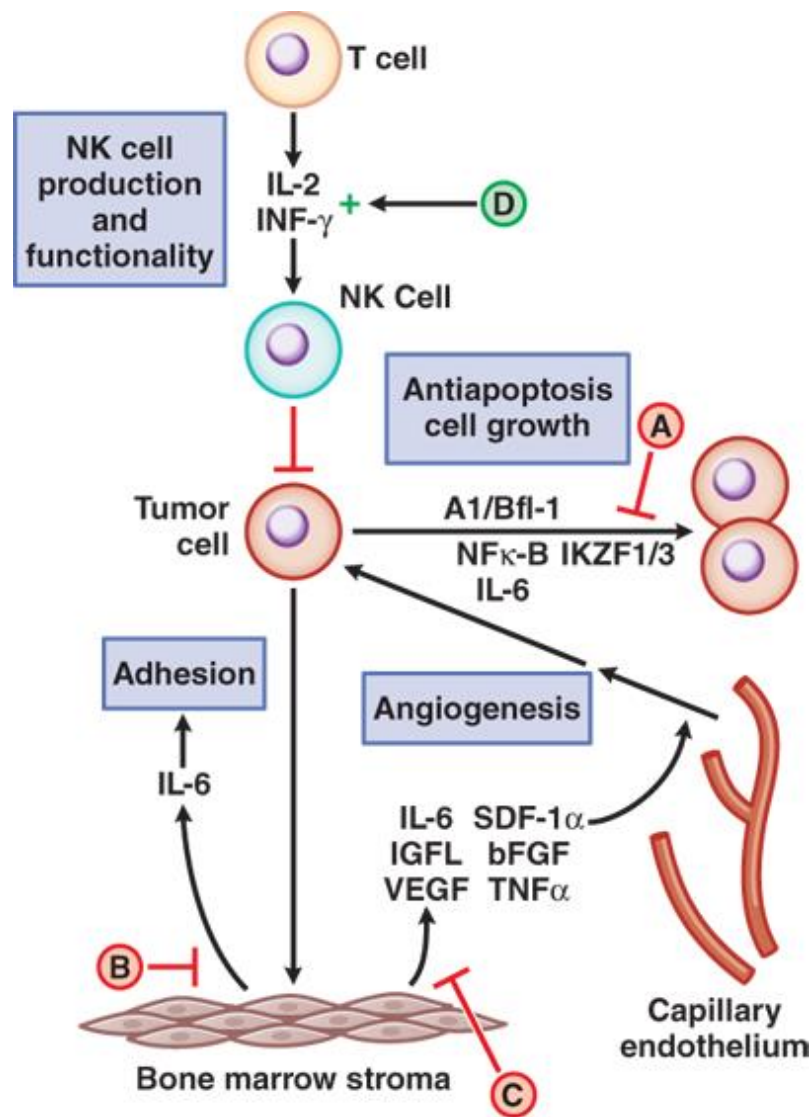
Ubiquitination (ubiquitylation): Post-translational modification in which ubiquitin is attached to a target protein. It is a tightly regulated, highly specific ATP-dependent biological process carried out by a cascade of enzymes.

Ubiquitination tags proteins for degradation, for subcellular localization, and for kinase activation.

E3 ubiquitin-protein ligases: A large family of proteins that regulate turnover and activity of proteins with diverse functions.

E3 ubiquitin-protein ligases transfer ubiquitin from E2 ubiquitin-conjugating enzyme to a substrate protein. The target protein is degraded by the 26S proteasome.

Cereblon: CRBN is a substrate adaptor the E3 ligase complex. CRBN forms an E3 ubiquitin-protein ligase complex with damaged DNA binding protein 1 (DDB1), Cullin-4A (CUL4A), and regulator of cullins 2 (ROC1). This complex ubiquitinates a number of other proteins.



Source: Laurence L. Brunton, Björn C. Knollmann:
 Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e:
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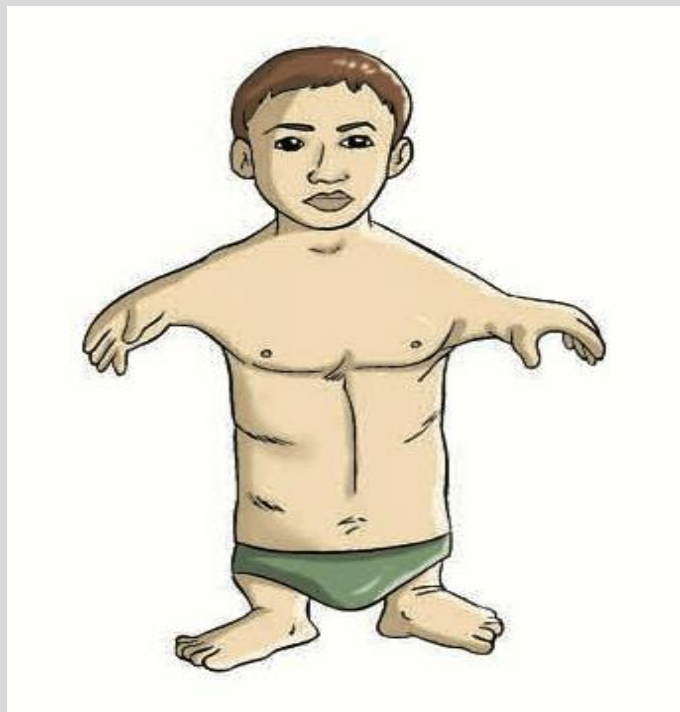
IMiDs immunomodulatory, antiangiogenic and antineoplastic actions.

- Direct antitumor effects (*ie* G1 arrest or apoptosis) by disrupting the antiapoptotic effect of BCL2 family members, blocking NF- κ B signaling, and inhibiting IL-6 production.
- Inhibit MM cell adhesion to bone marrow stromal cells partially due to the reduction of IL-6 release.
- Decrease angiogenesis due to inhibition of cytokine and growth factor production and release.
- Enhance cell-mediated cytotoxicity by increasing T cell production of cytokines, such as IL-2 and IFN- γ , that increase the number and cytotoxic functionality of NK cells.

Overview of proposed mechanisms of antitumor activity of thalidomide and its derivatives. Some biological hallmarks of the malignant phenotype are indicated in blue boxes. The proposed sites of action for thalidomide (letters inside red and green circles) are hypothesized to also be operative for thalidomide derivatives. A. Direct anti-MM effect on tumor cells, including G₁ growth arrest or apoptosis, even against MM cells resistant to conventional therapy. This is due to the disruption of the antiapoptotic effect of BCL2 family members, blocking NF- κ B signaling, and inhibition of the production of IL-6. Mechanistic studies show that lenalidomide can kill MM cells by inducing E3 ubiquitin ligase CRL4-CRBN-mediated ubiquitination and degradation of the crucial IKZF1 and -3 (zinc finger transcription factors). B. Inhibition of MM cell adhesion to bone marrow stromal cells partially due to the reduction of IL-6 release. C. Decreased angiogenesis due to the inhibition of cytokine and growth factor production and release. D. Enhanced T-cell production of cytokines, such as IL-2 and IFN- γ , that increase the number and cytotoxic functionality of NK cells.

Thalidomide: Pregnancy *Contraindicated*

- Embryotoxic: Severe developmental deformities including
- Stunted limb growth (phocomelia)
- Fetal cardiac, gastrointestinal, bone, external ear, eye, and genitourinary tract abnormalities have been described
- Mortality at or shortly after birth reported



[http://wordinfo.info/unit/3621/s:
a%20sea%20calf](http://wordinfo.info/unit/3621/s:a%20sea%20calf)



[http://www.math.lsa.umich.edu/mmss/2005/
/embryology/sabreim/Teratogens.htm](http://www.math.lsa.umich.edu/mmss/2005/embryology/sabreim/Teratogens.htm)

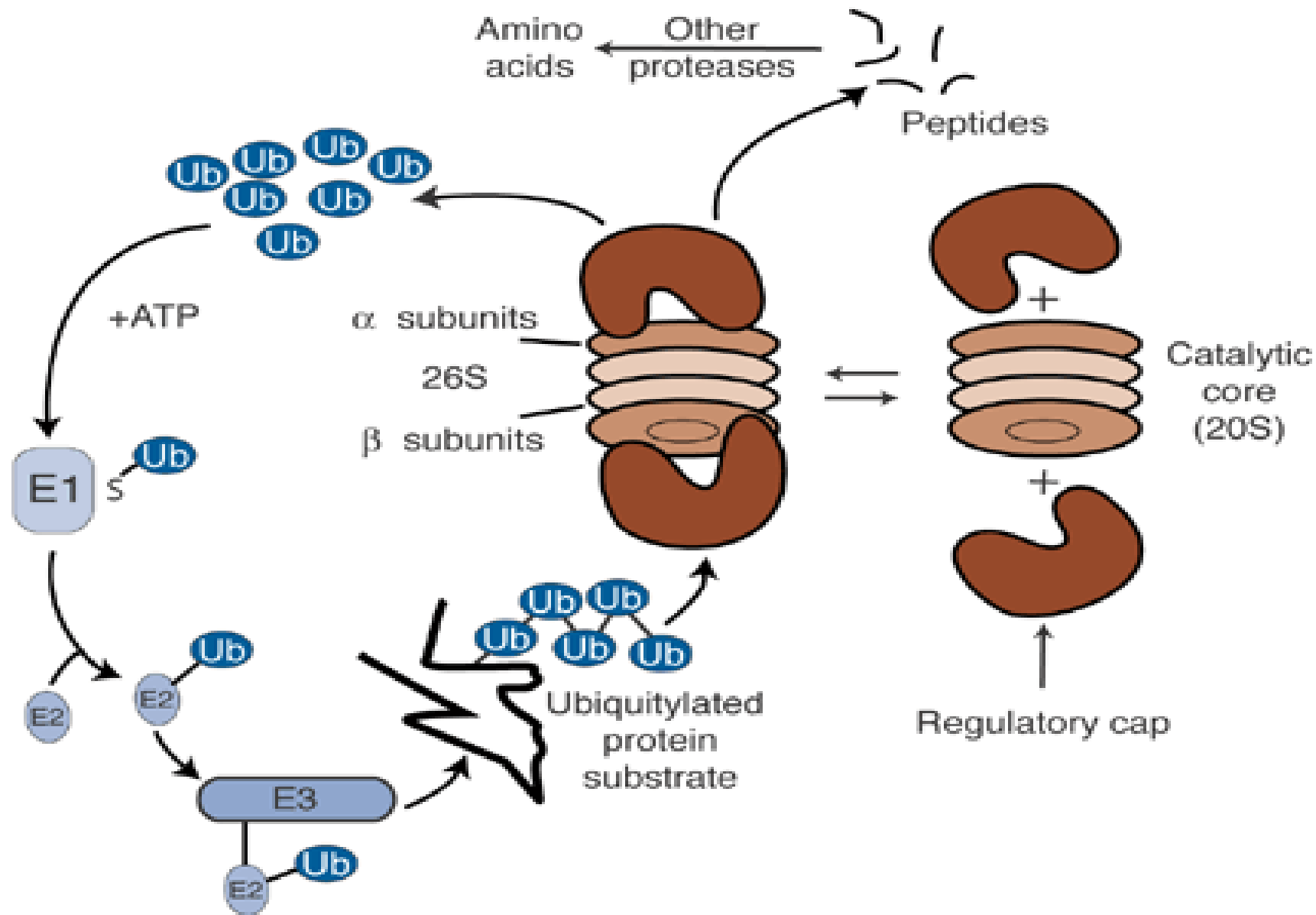
Thalidomide, lenalidomide, and pomalidomide are present in semen.

- Advise patients to use condoms with partners who could become pregnant during therapy and at least 4 weeks after treatment completion.

Pharmacology of Proteasome Inhibitors

	* Bortezomib	Carfilzomib	Ixazomib
PK	IV, hepatic CYPs	IV, peptidase	Oral, CYP3A4
MOA	Reversible	Irreversible	Reversible
Uses	MM induction MM relapsed/refractory Mantle cell lymphoma Cutaneous T cell lymphoma Peripheral T cell lymphoma Waldenstrom macroglobulinemia	MM relapsed/refractory Waldenstrom macroglobulinemia	MM relapsed/refractory
Toxicities	GI: nausea, vomiting, diarrhea, ileus Neutropenia and severe thrombocytopenia Peripheral neuropathy, sensory (stocking-glove) or mixed sensorimotor Heart failure, Hypotension Pulmonary toxicity (ARDS, pneumonitis, interstitial lung disease)		
	- The carfilzomib regimen (KRd) is associated with more dyspnea, heart failure, hypertension, and acute kidney injury compared to the bortezomib regimen. - The bortezomib regimen is associated with more peripheral neuropathy.		

Proteasome Inhibitors



26S Proteasome is a large protein complex of proteolytic enzymes that degrades ubiquinated (tagged) proteins, which is essential for homeostasis.

Proteasomes also selectively destroy potentially harmful abnormal or unfolded proteins, which might be produced in increased amounts in cancers.

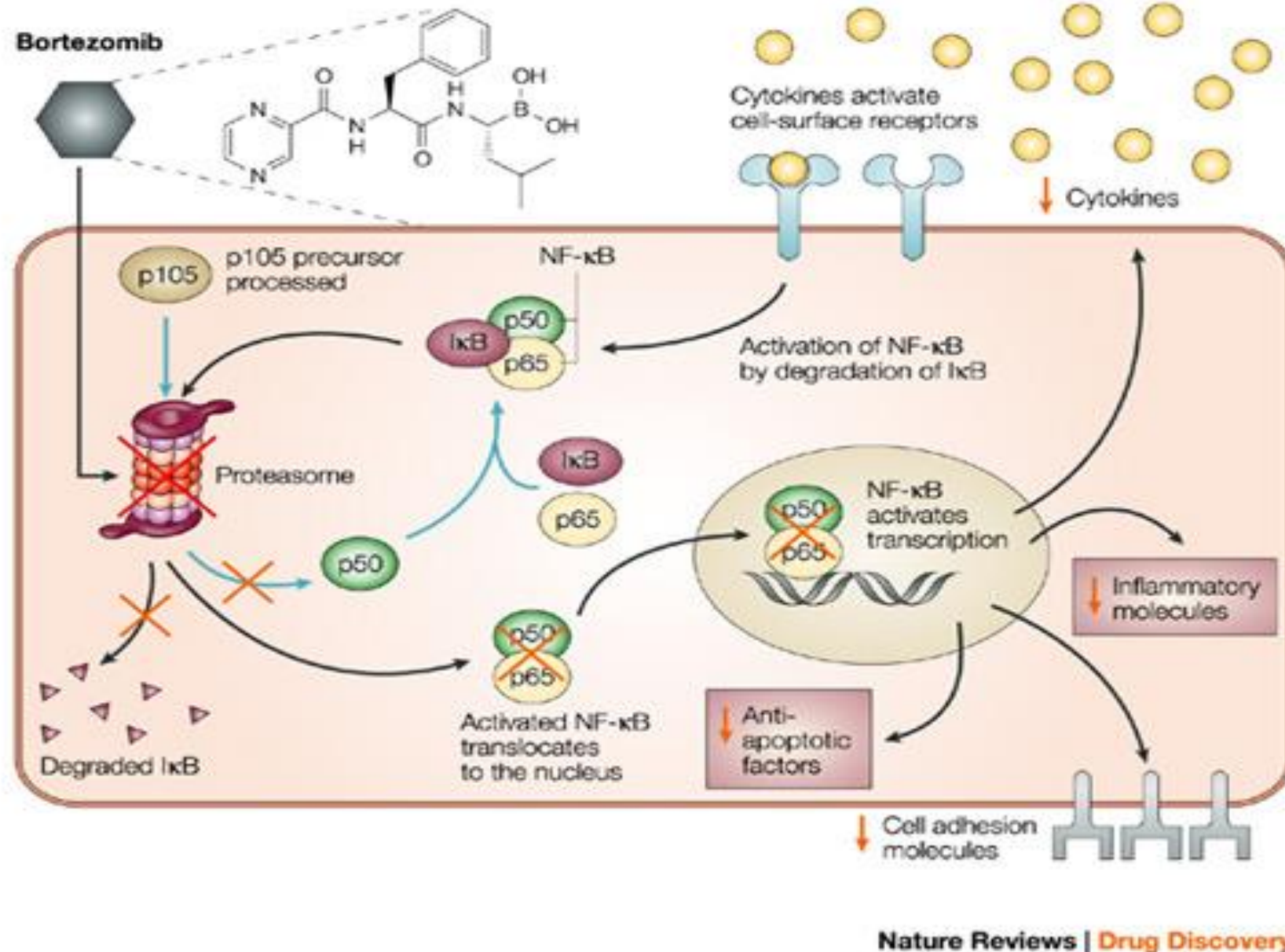
Amino acids and ubiquitin are recycled.

Nuclear factor- κ B (NF- κ B) is constitutively expressed in myeloma cells. The proteasome is essential for NF- κ B activation by releasing it from I κ B in the cytosol.

NF- κ B is essential for many tumor cell processes, including cell growth and survival.

Mechanism of action: Bind 20S core of 26S proteasome

Proteasome inhibitors induce apoptosis and prevent NF- κ B activation



drug molecule forms a complex
with the threonine hydroxyl group in the
chymotrypsin-like active site
↓
inhibits proteolysis of ubiquitinated
proteins
↓
accumulation of abnormal protein
conformations
↓
activation of signaling cascades induces
apoptosis
AND
proteasome inhibition prevents
degradation of I κ B, which prevents
NF- κ B activation

Anti-CD38 Agent: *Daratumumab

PK	IgG MAb I.V. Daratumumab-hyaluronidase SubQ $t_{1/2}$ 18 ± 9 days
MOA	CD38 is a cell surface glycoprotein highly expressed on myeloma cells but there are low levels of expression on normal lymphoid and myeloid cells. 1. Complement-dependent cytotoxicity 2. Antibody-dependent cellular cytotoxicity – ADCC (NK cells) 3. Fc-mediated crosslinking → apoptosis
Uses	MM, newly diagnosed induction regimen • High-risk, candidates for autologous hematopoietic cell transplant (HCT): • DVRd, VRd, or DRd followed by stem cell collection and early HCT MM, relapsed/refractory Systemic light chain amyloidosis (monoclonal plasma cell proliferative disorder)
Toxicity	Bone marrow toxicity Infusion reactions latent infection reactivation Screen for HBV infection Herpes zoster prophylaxis

Isatuximab is an anti-CD38 MAb for relapsed / refractory MM

Glucocorticoid: *Dexamethasone

- Dexamethasone is a component of 3-drug, 4-drug, and 2-drug regimens for initial treatment of multiple myeloma and relapsed and refractory multiple myeloma.
- Dexamethasone decreases inflammation by suppression of neutrophil migration, decreased production of inflammatory mediators, and reversal of increased capillary permeability; suppresses normal immune response.
- Dexamethasone induces apoptosis in multiple myeloma cells.

Check your knowledge:

You might want to pause the video to think about the answers.

- What is the initial therapy for standard-risk multiple myeloma?
- What drug is added for initial therapy in high-risk patients who are candidates for HCT ?
- What is the mechanism of action of each of the drugs in the regimen?
- What drugs are known to be embryotoxic and present in semen?

Cancer Immunotherapy

IMMUNE CHECKPOINT INHIBITORS

CTLA-4 Inhibitors

*Ipilimumab
Tremelimumab

PD-1 Inhibitors

*Nivolumab
Pembrolizumab
Cemiplimab
Dostarlimab

PD-L1 Inhibitors

*Atezolizumab
Avelumab
Durvalumab

T cell activation is tightly controlled by co-inhibitory molecules, immunosuppressive cells, and cytokines.

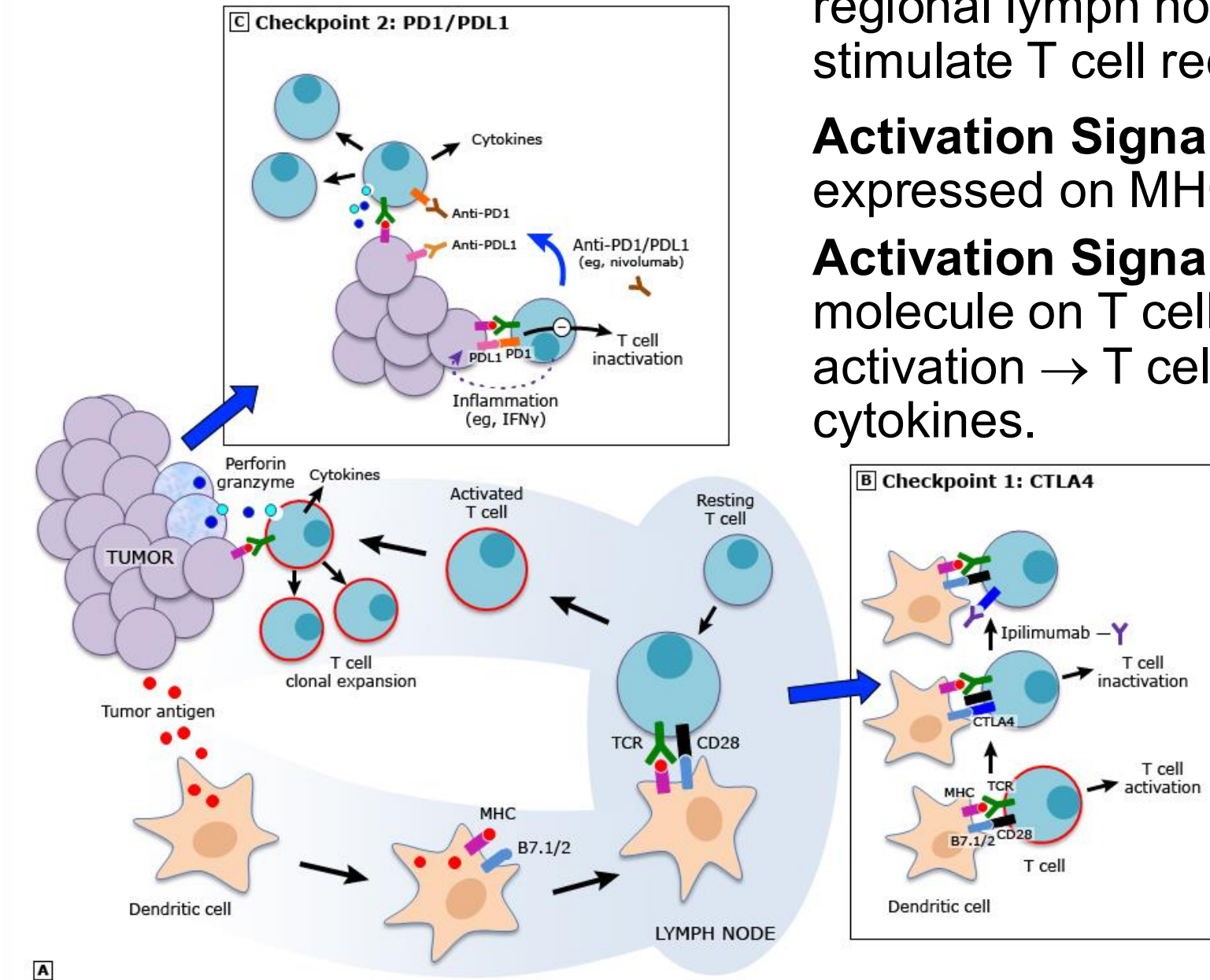
Immune checkpoints are inhibitory signals that limit T cell activation.

Malignancy can lead to immune tolerance of tumor and progression of malignancy.

Therapeutic approach:

Blocking immune checkpoints on cytotoxic (CD8+) T cells allows them to recognize cancer cells expressing antigens recognized by the T cells and eliminate them.

Immune activation and checkpoint inhibition



PD1: programmed cell death receptor 1; PDL1: programmed cell death ligand 1; IFN γ : interferon gamma; TCR: T cell receptor; CD28: cluster of differentiation 28; MHC: major histocompatibility complex; CTLA4: cytotoxic T-lymphocyte-associated protein 4.

Graphic 90346 version 4.0

Reproduced with permission from: Scott Gettinger, MD.

Antigens shed from dead cancer cells are captured and processed by antigen presenting cells (APCs). APCs travel to the regional lymph nodes and present the antigen bound to MHC to stimulate T cell receptors (TCRs).

Activation Signal 1: Engagement of T cells with antigen expressed on MHC of APCs

Activation Signal 2: Engagement of CD 28 co-stimulatory molecule on T cells with CD80/86 (B7.1/B7.2) on APC → T cell activation → T cell proliferation, migration, and production of cytokines.

Checkpoint 1: Expression of CTLA-4 co-inhibitory molecule on T cells → interrupts T cell activation by outcompeting CD28 for binding to CD80/86

Checkpoint 2: Expression of PD1, which binds PD-L1 → The PD-1:PD-L1 interaction directly inhibits tumor cell apoptosis, promotes peripheral T cell exhaustion, and increase in regulatory T cells (Tregs).

CTLA-4 Inhibitors:

Ipilimumab: Blocks CTLA-4 → allows for enhanced T-cell activation and proliferation

Uses: Melanoma, colorectal cancer, non-small lung cancer, and others in combination with chemotherapy or nivolumab

PD-1 Inhibitors:

(PD-1 is expressed on many cell types: T and B cells, NK cells, monocytes, dendritic cells.)

Block PD-1:PD-L1 interaction → release immune suppression → immune response against the tumor

Uses: Many metastatic solid tumors as monotherapy or in combination with cytotoxic chemo or other ICIs.

PD-L1 inhibitors:

(PD-L1 is expressed on tumor cells, NK cells, monocytes, dendritic cells.)

Block PD-L1 block → prevents the PD-ligand-mediated T cell suppression

Uses: Many metastatic unresectable solid tumors as monotherapy or in combination with cytotoxic chemotherapy or other ICIs.

Immune Checkpoint Inhibitors: Immune-related adverse events

Adverse effects are typically transient, but they can be severe or fatal due to T cell activation and proliferation.

Immune-related adverse events are more common with CTLA-4 inhibitors than with PD-1 or PD-L1 inhibitors.

- Inflammatory skin reactions, immunobullous disease, vasculitis, neutrophilic dermatosis: the most common adverse effects
 - GI: diarrhea, colitis
 - Hepatotoxicity: frequent, hepatocellular injury with asymptomatic elevations of ALT and AST; severe hepatotoxicity is rare
 - Endocrinopathies: Inflammation of the pituitary (hypophysitis), thyroid, or adrenal glands due to checkpoint blockade typically results in gland failure.
 - Pneumonitis: uncommon but can be severe and fatal
 - Autoimmune hemolytic anemia
 - Hemophagocytic lymphohistiocytosis (overactive lymphocytes and histiocytes)
- **Practice point:** Early recognition, timely intervention, including cessation of therapy, are critical. Treatment is based on the severity of the observed toxicity.

CAR-T Cell Therapies (“living drugs”)

NEW YORK INSTITUTE OF TECHNOLOGY

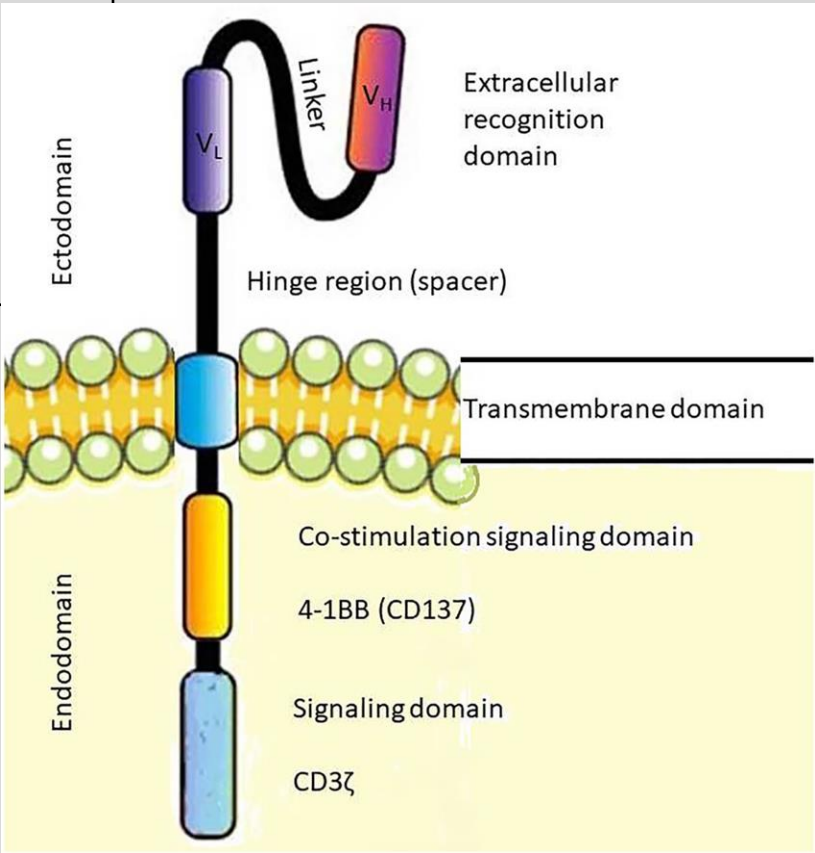
Ex vivo modification of autologous T cells are genetically engineered to direct the patient’s own T cells against their own tumor cells.

College of Osteopathic Medicine

The cells persist as memory CAR-T cells, providing ongoing antitumor actions.

CD19 targeting	BCMA targeting	MAGE-A4 targeting
*Tisagenlecleucel Axicabtagene ciloleucel Brexucabtagene autoleucel Lisocabtagene miraleucel	Idecabtagene vicleucel Ciltacabtagene autoleucel	Afamitresgene autoleucel

- T cells are harvested from the patient’s blood. The T cells are genetically engineered to express artificial T cell receptors targeting specific tumor antigens. The engineered T cells are returned to the patient’s body.
- Both antigen-binding and T cell activating functions are combined in one receptor. A murine CD19-binding antibody fragment that recognizes CD19 is fused to intracellular signaling domains from 4-1BB (CD137) and CD3-zeta.
- Anti-BCMA and anti-MAGE-A4 CAR-T cells are engineered in a similar way.



Drugs	Drug-specific indications	Toxicities
Anti-CD19 drugs	- B cell acute lymphoblastic leukemia - Diffuse large cell lymphoma - Follicular lymphoma - Mantle cell lymphoma	Cytokine release syndrome - Acute systemic inflammatory syndrome characterized by fever and multiple organ dysfunction - Onset: 1 to 22 days (clinical trials) - Monitor for at least 4 weeks after treatment - Hospitalize at the first sign: supportive care, tocilizumab and/or GC Immune effector cell neurotoxicity syndrome - Neuropsychiatric syndrome that occurs in patients treated with CAR-T therapy - ICANS of any severity in 20-70% of patients - Onset: days to weeks after admin of CAR T cell tx Encephalopathy with confusion, behavioral changes Visual and auditory hallucinations, language dysfunction, headache, tremors, seizures, cerebral edema, coma, death
Anti-BCMA Drugs BCMA: B cell maturation antigen	Multiple myeloma	
Anti-MAGE-A4	Synovial sarcoma, unresectable or metastatic, relapsed or refractory	

And many other adverse effects affecting all organ systems

Check your knowledge:

You might want to pause the video to think about the answers.

- What is the mechanistic concept of the immune checkpoint inhibitors (ICIs)?
- What types of cancers are the ICIs used for?
- What are some serious and potentially life-threatening toxicities of the ICIs?
- What is the mechanistic concept of the CAR-T cell drugs?
- What are some serious and potentially life-threatening toxicities of CAR T cells?

Summary

Targeting oncogenic pathways that contribute to cancer growth and metastatic spread as well as immune evasion

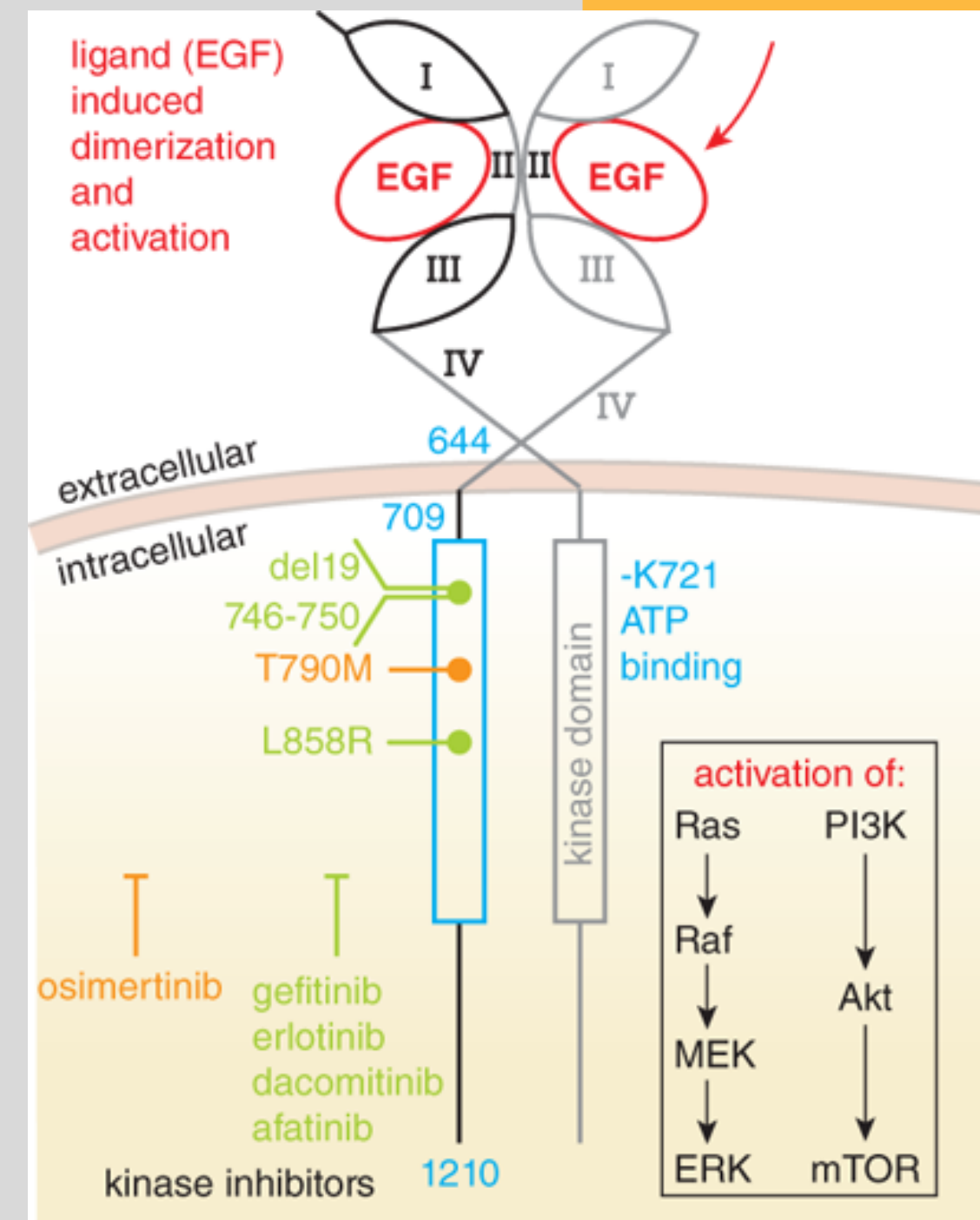
- Altered growth factor receptor signaling
- Activation of intracellular kinases
- Cancer cell-host interactions and aberrant angiogenesis
- Defects in DNA repair
- Altered protein degradation
- Altered epigenetic regulation
- Targets that control cancer cell behavior
- Inhibiting growth factors receptors in cancer cells
- Inhibiting tumor angiogenesis
- Activating immune cells
- Targeting cancer cell surface molecules to engage immune cells
- Cytotoxin conjugates with antibodies or cytokines
- Cellular therapy and vaccines

- Targeted anticancer agents have been developed for a wide array of cancers. They are generally used in combination regimens with cytotoxic chemotherapy or other targeted agents.
- Monoclonal antibodies bind to and inhibit specific receptors / cell surface proteins, which prevents dimerization and activation of the oncogenic pathway. Some MAbs induce antibody-dependent cellular cytotoxicity and complement-mediated cellular toxicity. Toxicities are related to inhibition of specific targets, such as receptor tyrosine kinases and vascular endothelial growth factor, infusion hypersensitivity reactions.
- Small molecular oral agents act at intracellular sites, such as protein kinases, preventing cellular functions such as gene expression, immune response, cell survival and division. Toxicities, such as dermatologic reactions and bone marrow suppression, are related to the inhibition of specific targets. The drugs are metabolized by hepatic CYPs, especially CYP3A4, and have a high potential for drug interactions.

- Immune checkpoint inhibitors are drugs that prevent the checkpoint proteins from downregulating T cell actions, thereby unleashing the T cell response against the tumor cells. Targets are CTLA-4, PD-1 and PD-L1. Toxicities include autoimmune reactions due to T cell activation and proliferation and infusion hypersensitivity reactions.
- CAR-T cells – chimeric antigen receptor T cells – are genetically engineered to express artificial T cell receptors targeting specific tumor antigens. Both the antigen-binding and T cell activating functions are combined in one receptor. CAR-T cells are used in the treatment of CD19-positive leukemias / lymphomas and BCMA-positive multiple myeloma. Potentially life-threatening toxicities include severe cytokine release syndrome, an inflammatory syndrome with fever and multiorgan failure, and immune effector cell-associated neurotoxicity syndrome (ICANS), which is a neuropsychiatric syndrome characterized by encephalopathy, hallucinations, and language dysfunction.

Summary of TKIs

- Many protein kinase inhibitors and pathway-targeted molecules have been developed for a wide variety of cancers expressing these oncogenic proteins.
- Two separate classes of drugs that target the EGFR pathway are important in the therapy of solid tumors: protein tyrosine kinase inhibitors (TKIs) and inhibitors of extracellular ligand binding.
- Monoclonal antibodies that recognize the extracellular domain of the EGFR, inhibit ligand-induced signaling. EGFR TKIs bind to the intracellular kinase domain and inhibit the enzymatic function of EGFR.
- The mechanisms and pharmacology are typical of the drugs described in this video for CML, colorectal cancer, and HER2-positive breast cancer.

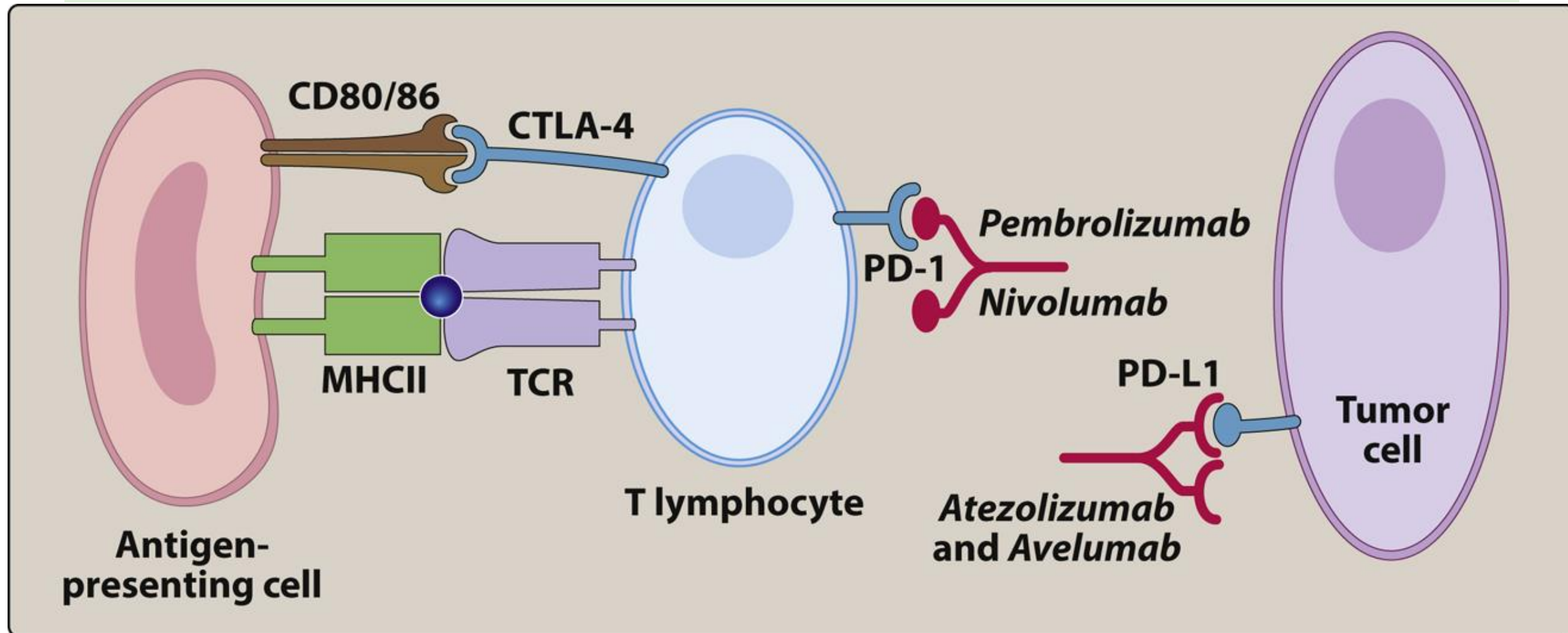


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Goodman and Gilman Figure 71-1
Targeting EGFR in cancer.

From: **37 Anticancer Drugs**

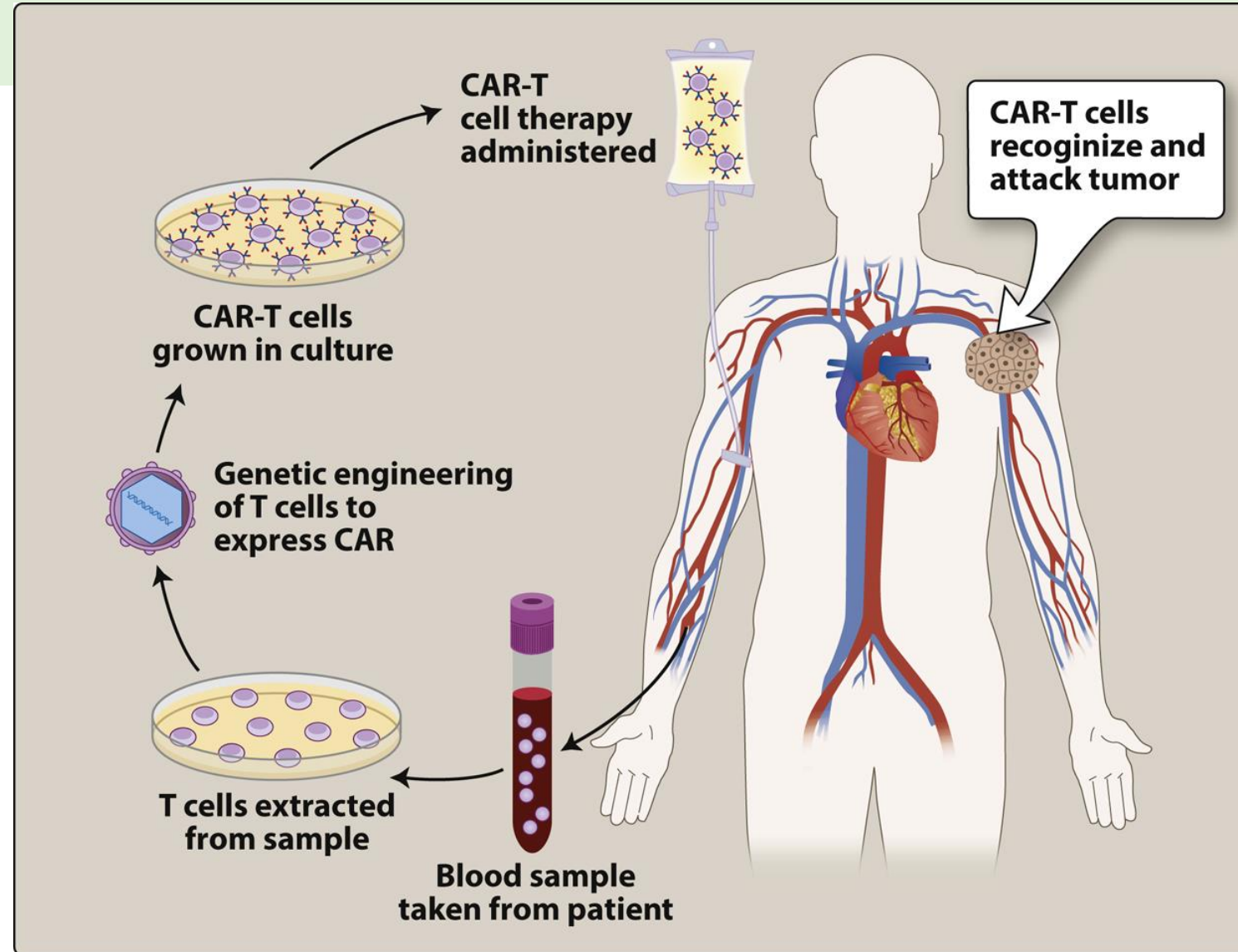
Lippincott® Illustrated Reviews: Pharmacology, 8e, 2023



Mechanism of action of immune checkpoint inhibitors.

From: **37 Anticancer Drugs**

Lippincott® Illustrated Reviews: Pharmacology, 8e, 2023



Gene and cell therapies for cancer.

Check your knowledge:

You might want to pause the video to think about the answers.

- What type of protein is expressed by the *BCR::ABL1* driver mutation?

Constitutively active cytosolic tyrosine kinase, BCR::ABL1.

- What is the importance of the T315I point mutation in the ABL1 contact residue?

This mutation confers resistance to all but two BCR::ABL1 inhibitors.

- What drugs are active against CML harboring this mutation?

Ponatinib and Asciminib

- What are some class toxicities of the tinibs group?

Diarrhea, rash, edema including orbital edema, bone marrow suppression, CYP3A4 drug interactions

- What are the serious and potentially life-threatening toxicities of ponatinib?

Arterial occlusive events → acute MI, stroke, brain arterial stenosis, PVD; heart failure; QT interval prolongation / arrhythmia; hepatotoxicity

Check your knowledge:

You might want to pause the video to think about the answers.

- Cetuximab and panitumumab are ineffective in tumors with what mutation?
KRAS, oncogene for a GTPase, a transducer of the RAF/MEK/ERK pathway
- What are the target and disease treated by both cetuximab and panitumumab?
Monoclonal antibodies directed against EGFR for metastatic colorectal cancer.
- What is the mechanism of bevacizumab?
Bevacizumab neutralizes soluble VEGF preventing it from activating its RTKs.
- What cytotoxic chemotherapies are included in the combination regimen?
Folinic acid (leucovorin) + 5-fluorouracil + oxaliplatin + cetuximab + bevacizumab
- What are some serious and potentially life-threatening toxicities of these MAbs?
EGFR inhibitors: diarrhea, rash, electrolyte disturbances, lung toxicity, infusion rx
VEGF inhibitor: hemorrhage, poor wound healing, hypertension, thromboses, GI perforation, infusion reactions

Check your knowledge:

You might want to pause the video to think about the answers.

- Mechanistically, how do trastuzumab and pertuzumab work together?

More complete inhibition of HER2 signaling by binding different epitopes in HER2.

- Mechanistically, how does lapatinib interfere with HER2 signaling?

Lapatinib is an oral small molecule that prevents activation of intracellular receptor tyrosine kinases, which prevents activation of second messengers.

- What is the mechanistic concept of antibody-drug conjugates?

The antibody directed against a specific target delivers a cytotoxic payload to those cells expressing the receptors, thereby limiting exposure of host cells.

- What cytotoxic chemotherapies are included in the combination regimen?

A taxane and/or doxorubicin, cyclophosphamide, trastuzumab, $\pm$ pertuzumab

- What are some serious and potentially life-threatening toxicities of trastuzumab?

Heart failure (esp. w. doxorubicin), lung tox, infusion reactions,

Check your knowledge:

You might want to pause the video to think about the answers.

- Rituximab targets what antigen on what types of immune cells?

CD20 cell surface protein expressed on 95% of B cells in peripheral blood and secondary lymphoid organs. CD20 is absent on terminally differentiated plasma cells.

- What is rituximab's mechanism of tumor and autoimmune cell eradication?

Rituximab induces antibody-dependent cellular cytotoxicity and complement-dependent cytotoxicity.

- Patients should be screened for what infections prior to beginning rituximab tx?

All patients should be screened for HBV infection. In high-risk populations, screen for latent infections, including tuberculosis, hepatitis C virus, and HIV.

- What is the purpose of administering rituximab in conjunction with hyaluronidase?

Hyaluronidase increases permeability of subcutaneous tissue through temporary depolymerization of hyaluronan (high mw polysaccharide in extracellular matrix), which increases the dispersion and absorption rate of the drug.

- How do CDK4/6 inhibitors and PARP inhibitors cause tumor cell death?

CDK4/6 inhibitors prevent release of Rb tumor suppressor protein, which prevents progression from G1 to S cell cycle phase, leading to cell cycle arrest. PARP inhibitors block DNA damage response (DDR) pathways in patients with deleterious BRCA1/2 tumors, which prevents tumor cells from repairing DNA, making them vulnerable to apoptosis.

Check your knowledge:

You might want to pause the video to think about the answers.

- What is the initial therapy for standard-risk multiple myeloma?

VRd: Bortezomib, lenalidomide, dexamethasone

- What drug is added for initial therapy in high-risk patients who are candidates for HCT ?

Daratumumab, CD38 inhibitor

- What is the mechanism of action of each of the drugs in the regimen?

Proteasome inhibitor prevents proteolysis of ubiquitinated proteins and release of NF- κ B. IMiDs recruit substrates, resulting in ubiquitination and degradation of transcription factors. Dexamethasone induces apoptosis of MM cells. Daratumumab induces ADCC, complement-dependent cellular cytotoxicity, and apoptosis.

- What drugs are known to be embryotoxic and present in semen?

Immunomodulatory imide drugs IMiDs.

Check your knowledge:

You might want to pause the video to think about the answers.

- What is the mechanistic concept of the immune checkpoint inhibitors (ICIs)?

Immune checkpoints are inhibitory signals that limit T cell activation. Blocking immune checkpoints on cytotoxic (CD8+) T cells allows them to recognize cancer cells expressing antigens recognized by the T cells and eliminate them.

- What types of cancers are the ICIs used for?

Many types of solid tumors.

- What are some serious and potentially life-threatening toxicities of the ICIs?

Immune-related adverse reactions and infusion reactions.

- What is the mechanistic concept of the CAR-T cell drugs?

CAR-T cells are genetically engineered to express artificial T cell receptors targeting specific tumor antigens. Both the antigen-binding and T cell activating functions are combined in one receptor.

- What are some serious and potentially life-threatening toxicities of CAR T cells?

Cytokine release syndrome and immune effector cell neurotoxicity syndrome



All photographs by Aleya Cydney Photography from a Brushes With Cancer event in Chicago.

<https://www.phillymag.com/be-well-philly/2019/09/05/cancer-art/>

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